

aTOMos v3.6.1.14

Unified field formalization

Exact operator seeds

Coupled physical fields and measured outputs

Literal ASA/NA and JK state feedback

Electromagnetism, mechanics, heat, gravity and time

One-bit words, energy balances and calibration

Tom Klootwijk

Author-supplied identifier	NL200678942
Author-supplied date	10-07-1990
Release	3.6.1.14 / 15 September 2026
Underlying word profile	XOP-R1 / XOPSEED1 / XOPPLAN1
Method	Logical derivation and primary-source review

A connected mathematical framework: the seed carries the equations, the physical model supplies their meaning, shared energy determines coupling, and the observation model determines what an output can establish.

Established physics is integrated under explicit model assumptions. “Unified” describes the field/state framework; no new fundamental force-unification theory is asserted. Prepared with AI assistance.

One formalization, connected by explicit equations

This release integrates the exact seed, literal state feedback, physical field models and measurement contract in one document. Its contribution is their explicit mathematical composition. Maxwell's equations, classical mechanics, thermodynamics and weak-field gravity retain their established scientific provenance. Material response, finite reductions and instrument parameters are selected models whose applicability must be established for a device.

The central construction uses a common total energy for finite electromagnetic, polarization, mechanical and thermal states. Reciprocal interactions follow from its derivatives. Dissipation transfers energy into the declared thermal stores exactly once. A separate finite quadratic-field step derives its own discrete work and heat identity. The gravity/time profile connects acceleration, clock rate, source time and ground-station observation under one convention. These mathematical bodies become typed XOP expressions inside the same causal ASA/NA and JK architecture; a separate display does not overwrite exact roots.

Layer	Included mathematical content	Connection to the next layer
Representation	One-bit arithmetic, exact values, typed expressions, complete wire grammar	Stores actual operators and their dependencies
State feedback	Two ordered ASA/NA sets, literal JK and delayed samples	Selects commands and observations with defined causality
Physical laws	Field, momentum, energy, entropy and gravity/time equations	Constrains the selected constitutive and spatial models
Common coupling	Shared energy gradients and explicit power/heat exchange	Prevents inconsistent forces or duplicated loss
Finite evolution	Exact linear solves and derived discrete balances	Specifies the actual mathematical step retained in the seed
Observation	Acquisition, clocks, calibration, uncertainty and tolerance	Defines the claim attached to an exported physical result

The result is a reusable foundation for digital, physical, analog, photonic, spatial and calibration specializations. It is not an automatic selection of the right physics for every application. Each specialization closes its own inputs, boundaries, admissible domain and requested output. A finite continuum reduction remains a model even when its retained arithmetic is exact.

Lineage and present scope

R11 supplied the exact expression and word contract, R12 the WANTWOMBAN photonic and sensing interfaces, and R13 the physical-law foundation. R14 restates the exact core and full wire/operator catalog, integrates the physical chapters and adds the common coupling, field step and gravity/time connections. This is a consolidated foundation, rather than an appended copy of the older application manuals or historical measurement tables. The named older runtime profiles and their original evidence remain in their preserved releases.

The current profile is XOP-UNIFIED-FIELD-R1. Its finite templates use the unchanged XOP-R1 vocabulary; no new opcode is implied by a physics equation. The body and appendix contain the representation and shared physical contracts. A referenced optional packet transport or historical application remains a separately versioned dependency when selected.

Continuous fields remain equations

The authoritative field may be a function of continuous space and time. For example, a slit in a declared plane has the literal predicate

$$A(x) = \mathbf{1}\{|a \cdot (x - x_c)| \leq w/2\} \mathbf{1}\{|b \cdot (x - x_c)| \leq h/2\}, \quad a \cdot b = 0, \quad \|a\| = \|b\| = 1,$$

with exact parameter expressions and explicit physical uncertainty. Its definition has no pixels. A function, integral or ODE solution can likewise retain exact denotation without rasterizing its domain. The XOP catalog gives the corresponding domain conditions; a retained solution term is not a claim that a finite executor has evaluated every real value. Selecting a modal truncation or finite time step is an explicit application choice. A display sampling never becomes the authoritative field merely because it was rendered.

Evidence and authorship

The work comprises equation derivation, primary-source review, internal independent cross-review and document inspection. No kernel/reference programs, numerical trials, algorithm tests, simulations or benchmarks were executed for this release. At the user's request, a real NIST internet time reply was acquired and latched as immutable source data, with its raw timestamp, protocol metadata and dated time-scale conversion. This observation provides an atomic-service reference, not a local atomic oscillator or a physical accuracy experiment on the proposed kernels. Compiling the manuscript and checking artifact bytes establish document delivery and provenance. They establish no new runtime, physical implementation, satellite accuracy or manufactured photonic performance.

Author attribution is preserved as supplied by Tom Klootwijk. External laws and references are attributed near their equations. The aTOMos composition, selected model definitions and conditional derivations are identified as such. Editable source and provenance accompany the PDF, with the final PDF also placed directly in the main Git repository.

Contents

One formalization, connected by explicit equations	2
1 Exact operator seeds: a mathematical contract	5
2 The exact kernel, physical fields and acquired observations	12
3 A minimal physical-law layer for application kernels	16
4 An exact finite step with a derived mechanical energy balance	18
5 Momentum, energy and admissible physical coupling	20
6 Electromagnetic closure of the optical interface	25
7 One coupled field state and one physical energy	29
8 A finite exact field kernel with work and heat balance	33
9 Gravity, clocks and station observations from one state	35
10 An atomic-time observation, literal latch and domain bindings	38
11 Measurement, constants and output tolerances	41
12 Specialization, proof scope and delivered authority	45
A Normative XOP-R1 grammar and operator registry	47

1 Exact operator seeds: a mathematical contract

This chapter restates the XOP-R1 operator-seed contract introduced in release 3.6.1.11 and adopted by the unified field profile. Definitions specify a representation and its meaning; propositions establish conditional mathematical consequences. An implementation must still discharge the stated parsing, arithmetic and proof-checking obligations. No new runtime, machine-checked proof, test, simulation or measured improvement is asserted. The implemented R10 orbital seed and its binary64 arithmetic keep their existing format and interpretation.

1.1 Bits, values and exact expressions are distinct types

Write $\text{Bits}(w) = \{0,1\}^w$ for a finite raw bit string. Its unsigned interpretation is an explicitly requested map

$$\text{uint}(b) = \sum_{i=0}^{w-1} b_i 2^i.$$

A binary64 payload belongs to $\text{Bits}(64)$ until an explicit conversion is applied. It does not become an arbitrary-precision real by being packed in one-bit planes. A finite binary64 value can be converted to the exact dyadic rational denoted by that value; this preserves its rounded input value, not an unknown physical quantity. Signed zeros are distinct raw payloads but the same rational zero. Infinity and NaN have no value in the exact real scalar types.

The scalar hierarchy is

$$\mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{A} \hookrightarrow \text{RealTerm}, \quad \mathbb{A} = \overline{\mathbb{Q}} \cap \mathbb{R}.$$

Here RealTerm is finite syntax for a possibly partial real expression, not a claim that every real number has a finite description. Coercions preserve denotation and are explicit in the typed graph. Booleans, finite words, tuples, vectors, matrices, functions and streams have separate type constructors. Dimensioned scalars also carry a seven-component rational SI exponent vector; addition requires equal dimensions, multiplication adds dimensions, and transcendental functions require dimensionless arguments. Angle conventions and coordinate frames are explicit semantic annotations, not inferred from an untyped number. A framed vector's type carries its fully qualified nominal frame identity; a framed matrix carries source and target identities. Each identity contains namespace, major/minor version and label. The full annotation is serialized and participates in sharing. A frame change requires an explicit typed template and its coordinate-realization obligations, not a relabelled vector. Bare mathematical vectors carry no implied physical frame.

An integer is $(s, a_0, \dots, a_{k-1})$ with $s \in \{0,1\}$ and

$$z = (-1)^s \sum_{i=0}^{k-1} a_i \beta^i, \quad \beta = 2^{64}, \quad 0 \leq a_i < \beta.$$

Zero has $s = 0, k = 0$; otherwise $a_{k-1} \neq 0$. There is no format-level bound on k . A resource-limited implementation must return an explicit resource result, never silently discard high limbs. A rational is (p, q) with $q > 0$, $\gcd(|p|, q) = 1$; its zero is $(0, 1)$.

An algebraic real has a primitive irreducible polynomial $P \in \mathbb{Z}[X]$ with positive leading coefficient and an index selecting one of its distinct real roots in increasing order. This fixes its value and canonical mathematical identity. A rational isolating interval is a separate witness, not part of that identity. Checking irreducibility, isolating roots and obtaining a canonical minimal polynomial are explicit exact-algebra obligations. A general operator term has canonical *syntax*; different syntax may denote the same real.

1.2 Exact arithmetic on finite machine words

The literal full adder takes three bits $a, b, c \in \{0, 1\}$ and returns

$$s = a \oplus b \oplus c, \quad c' = (a \wedge b) \vee (c \wedge (a \oplus b)), \quad a + b + c = s + 2c'.$$

For subtraction with incoming borrow d , its bit and outgoing borrow are

$$r = a \oplus b \oplus d, \quad d' = (\neg a \wedge b) \vee (d \wedge \neg(a \oplus b)), \quad a - b - d = r - 2d'.$$

Feed the outgoing carry or borrow of position i to position $i + 1$, starting with zero. Signed values are handled by separate sign and magnitude, not by discarding an infinite sign extension. A product begins with partial bits $p_{i,j} = a_i \wedge b_j$ in position $i + j$; finitely many full-adder chains sum these shifted rows, retaining the final carry. Exact magnitude comparison scans from the most significant padded bit, with equality flag $e = 1$ and greater flag $g = 0$, updating simultaneously

$$g' = g \vee (e \wedge a \wedge \neg b), \quad e' = e \wedge \neg(a \oplus b).$$

This comparison supplies the branch in division; no approximate sign oracle is needed for integers. The bit equations also establish the limb implementation: $a + b = (a \oplus b) + 2(a \wedge b)$ and the two carry terms are disjoint, giving the full-adder identity by substitution. Summing the bit identities with powers of two telescopes all internal carries and borrows.

Limb storage does not force modular arithmetic. For nonnegative addition, pad the shorter operand with zero limbs, set $c_0 = 0$, and define

$$u_i = a_i + b_i + c_i, \quad r_i = u_i - \beta \lfloor u_i / \beta \rfloor, \quad c_{i+1} = \lfloor u_i / \beta \rfloor.$$

Append the final nonzero carry. A native implementation may realize u_i with two-word operations or bitwise carry logic; it cannot assume that u_i fits one limb. Subtraction with $A \geq B$ uses $c_0 = 0$ and

$$u_i = a_i - b_i - c_i, \quad c_{i+1} = \lfloor u_i < 0 \rfloor, \quad r_i = u_i + \beta c_{i+1}.$$

Signed addition selects addition or magnitude subtraction according to signs and exact magnitude order, then normalizes zero.

Multiplication is finite convolution followed by carrying, starting with $c_0 = 0$:

$$v_j = c_j + \sum_{i+\ell=j} a_i b_\ell, \quad r_j = v_j \bmod \beta, \quad c_{j+1} = \lfloor v_j / \beta \rfloor.$$

Continue carrying until it vanishes. The accumulator v_j is itself an exact multiword quantity; using one overflowing hardware word would violate the rule. For division by $D > 0$, process dividend bits from most to least significant:

$$Q_0 = R_0 = 0, \quad u_i = 2R_i + b_i, \quad e_i = \lfloor u_i \geq D \rfloor, \quad R_{i+1} = u_i - e_i D, \quad Q_{i+1} = 2Q_i + e_i.$$

The dividend prefix processed after step i is $DQ_i + R_i$, with $0 \leq R_i < D$. Signed Euclidean division is the unique $z = dq + r$ with $d \neq 0$ and $0 \leq r < |d|$; sign conversion must preserve this convention, including negative dividends. Integer powers with nonnegative exponent use finite repeated multiplication; negative powers produce a rational only when the base is nonzero.

Proposition 1.1 (No mathematical fixed-width overflow). *For finite integer operands, the preceding finite procedures return their exact sum, difference, product, or Euclidean quotient and remainder, provided every intermediate carry is retained and sufficient storage is available.*

Proof. For addition and subtraction, multiply each defining equation by β^i and sum: the carry terms telescope, leaving the represented operand relation. Convolution represents the product by distributivity, and carrying preserves each coefficient's value. For division, the prefix invariant follows by substitution, while $0 \leq R_i < D$ implies $0 \leq 2R_i + b_i < 2D$ and hence preserves the remainder bound. These arguments are conditional on retaining every term; they establish no bound on time or allocated memory. $\square$

1.3 Positive roots and elementary functions

$\text{SQRT}(x)$ denotes the unique $y \geq 0$ satisfying $y^2 = x$, with domain $x \geq 0$. $\text{ROOT}(n, x)$ requires integer $n \geq 1$: for even n , it is the nonnegative root and requires $x \geq 0$; for odd n , it is the unique real root. An arbitrary polynomial root uses the algebraic root index above, or an explicit existence-and-uniqueness predicate for a real-term root. No unnamed branch is selected. Thus $\sqrt{x^2} = |x|$ for real x , and cancellation to x requires $x \geq 0$. Complex branches are outside this real profile.

The real elementary declarations have ordinary mathematical denotations:

$$\begin{aligned} \exp x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, & \log x &= \int_1^x \frac{du}{u} \quad (x > 0), \\ \text{atan } x &= \int_0^x \frac{du}{1+u^2}, & \pi &= 4 \text{atan}(1). \end{aligned}$$

These definitions permit finite operator terms for infinite-process values. They are not a request to store an infinite series or an assertion that a finite limb evaluator can return all digits. A certified finite approximation needs separate truncation and rounding bounds. A raw approximate evaluation cannot serve as an exact equality proof.

$\text{ATAN2}(y, x)$ is undefined at $(0, 0)$ and uses range $(-\pi, \pi]$:

$$\text{atan2}(y, x) = \begin{cases} \text{atan}(y/x), & x > 0, \\ \text{atan}(y/x) + \pi, & x < 0, y \geq 0, \\ \text{atan}(y/x) - \pi, & x < 0, y < 0, \\ \pi/2, & x = 0, y > 0, \\ -\pi/2, & x = 0, y < 0. \end{cases}$$

The literal source OTAN2 remains a distinct named template with its own equation and domain; it is never an alias silently replaced by this operator. $\tan x$ requires $\cos x \neq 0$; inverse sine and cosine use ranges $[-\pi/2, \pi/2]$ and $[0, \pi]$ and domain $[-1, 1]$. Floor is the unique integer n with $n \leq x < n + 1$; wrapping is $x - P \lfloor x/P \rfloor$ for $P > 0$, with the integer winding retained separately.

1.4 Decisions, partiality and honest result states

Exact rational order is decided by integer cross multiplication with positive denominators. Algebraic order and equality can be decided by exact polynomial arithmetic, common-root analysis and real-root isolation. For unequal algebraic roots, sufficiently refined disjoint rational isolating intervals determine the order; common roots are handled algebraically instead of waiting forever for interval separation. Supplying those algorithms is an implementation obligation.

There is no general equality or sign oracle for unrestricted transcendental terms. COMPARE returns one of LT, EQ, GT, or UNKNOWN, each decided result carrying a valid justification. FLOOR , branch selection and nonzero checks may likewise remain unresolved. Outcomes distinguish a value or a retained term, an unmet domain obligation, an unknown decision, and exhausted resources. These are not numeric zero. IF is lazy: only the selected branch must be defined once its predicate is proved; an undecided predicate preserves the conditional term.

An exact finite decimal input denotes a rational with its declared decimal scale. A measurement with error is a value plus an explicit uncertainty model; giving its central value an exact rational spelling removes no measurement error. This distinction applies equally to station coordinates, timing and force coefficients in an orbital seed.

1.5 Scoped operator graphs and named dependencies

An operator seed contains finite typed records. A node is

(opcode, result type, ordered operand references, opcode payload).

The fixed XOP-R1 catalog in `formal/OPERATOR_CATALOG.md` supplies each opcode's arity, type relation, denotation, domain and payload fields. An unknown opcode is rejected; a user-defined formula is a named template built from catalog operators, with an explicit interface and body. There is no implicit "evaluate arbitrary text" opcode or undeclared exact solver.

Each scope has an ordered node table and ordered exported references. Local node operands refer only to earlier nodes. Lambda, finite sum/product, integral, derivative and solution declarations carry nested scopes. Bound variables use de Bruijn indices: zero denotes the nearest binder, and larger indices count outward through lexical binders. The type environment fixes every index. Substitution shifts indices when passing a binder and is capture-avoiding. Nested scopes cannot reference an outer local node by an accidental numeric ID; captured values enter through explicit binders. Names are display metadata; renaming them does not change variable binding.

References have distinct tags for a local node, bound variable, old state slot, current sample value, dependency export, raw sample timestamp, or converted GPST timestamp. A template application additionally identifies its template and supplies all parameters in declared order. Dependency records carry the complete canonical seed/template bytes in the self-contained form. A digest is a lookup aid; acceptance requires the expected complete bytes and interface. A dependency-only manifest is explicitly an open package until all required bytes are supplied. Dependency and template-call graphs are finite and acyclic in this profile. Finite summation and delayed state do not imply unrestricted recursive template calls.

Sample declarations identify their type, absolute time convention, provenance and binding mode. A time convention has a versioned semantic profile identity; the registered GPST convention supplies its stated scale and epoch, while a custom convention includes a complete dependency/template mapping and its domain. A name alone does not define a clock, resolve a leap second or prove monotonicity. An inline sample is an indexed literal accompanied by its absolute timestamp; an external sample is a declared input requirement. `SAMPLE` selects the value bound to the current logical step, with its timestamp retained. Missing data is unresolved input, not a guessed next sample. Raw-time reference tag 5 returns the sample's exact rational in its declared source-clock coordinate. Converted-time tag 6 explicitly applies the versioned mapping and returns GPST SI seconds; the value and both timestamp references read one immutable sample snapshot. Integration arguments are explicit expressions such as $t = \tau_{\text{GPST}} - \tau_{\text{epoch}}$ and $h = \tau_{\text{GPST}} - \tau_{\text{previous}}$, with matching time units and epoch convention. A custom raw clock tick is not implicitly a second, and a missing mapping cannot be replaced by an assumed offset. A modular phase, spatial key, hash or template identifier does not supply the state, timestamp or sample bytes it refers to.

1.6 Canonical variable-length wire contract

The canonical semantic stream begins with the eight ASCII octets `XOPSEED1`, followed by profile numbers (1, 0) and the ordered tables specified in the catalog. Every natural number uses minimal unsigned base-128 encoding: digit payloads occupy seven low bits, the high bit marks continuation, the least significant digit comes first, and the final high bit is zero. Zero is one zero octet; a multi-octet number cannot have a zero final payload. Thus counts and references are not limited to a fixed machine word.

A byte string has a natural-number length followed by exactly that many octets; a sequence has an element count followed by exactly that many schema-defined elements. An integer has a sign octet, a limb count and that many little-endian 64-bit limbs. Rational and algebraic literals use

the canonical value forms already defined. Strings used as identifiers are restricted ASCII byte strings; free-form provenance is opaque bytes and does not participate in name lookup. Types, references and opcode payloads are encoded by their declared tagged schemas. Nonminimal counts, redundant leading limbs, trailing bytes, nonzero padding, invalid references and payload fields outside the selected schema have no canonical decoding.

Canonical here means one encoding for the same ordered, scoped syntactic record. It does not mean that all mathematically equal expressions, all sharing patterns, or all valid topological node orders receive one universal byte string. Structural sharing is permitted only for identical typed syntax under the same bindings and immutable dependency identities. The strategy history records any rewrite that changes the syntax while preserving its meaning.

Proposition 1.2 (Unique literal and record decoding). *Given the catalog, a finite well-formed canonical stream has one decoded record and its canonical re-encoding returns the original stream.*

Proof. The continuation terminator uniquely delimits each minimal natural-number encoding. Lengths and fixed-width limbs uniquely delimit strings and literals. The record schema fixes field order and the selected tag fixes each variant. Induction over finite sequences and nested scopes therefore determines every field and its boundary. Rejecting alternative zero forms, redundant lengths and trailing bytes makes re-encoding identical. This is a structural proof of the specified grammar, not evidence about a parser implementation. $\square$

To pack literal one-bit planes, let C be the canonical octet stream, of length L . Pad its last block with zero octets to a multiple of 512 octets. Interpret each block as 64 little-endian words $w_0, \dots, w_{63}$ and set

$$a_{r,c} = ((w_r \gg c) \& 1), \quad p_c = \sum_{r=0}^{63} a_{r,c} 2^r.$$

The envelope is eight ASCII octets XOPPLAN1, then the minimal encoding of L , then the plane words in increasing block and plane order. Its size fixes the required block count; after inverse transpose, padding must be zero and the first L octets must decode as XOPSEED1. Empty input has no seed grammar, although the underlying byte permutation can act on an empty byte string.

Proposition 1.3 (Bit-plane involution and information accounting). *The padded block transform T satisfies $T(T(W)) = W$. Including L makes packing injective on canonical seeds. The transform changes organization, not information content or the number of bits required by its uncompressed padded blocks.*

Proof. The output bit at (c, r) is the input bit at (r, c) . Transposing again restores every coordinate. The explicit length removes padding ambiguity. Consequently distinct canonical inputs cannot share an envelope. A bijective permutation cannot itself identify multiple arbitrary inputs with a smaller code space. $\square$

This envelope is newly named. It neither changes nor retroactively reinterprets R10's uint64/binary64 codec. A conversion from R10 must declare whether it preserves raw bits, converts finite values to dyadic rationals, or replaces an approximation by a separately justified symbolic model.

1.7 Causal recurrence and the literal two-mask kernel

The state table declares each slot's type, initial expression and next-state expression. Reading slot j at logical step t is precisely

$$s_j(0) = \text{init}_j, \quad s_j(t+1) = E_j(s(t), d(t)); \quad s_j = \text{DELAY}(\text{init}_j, E_j).$$

All right-hand sides read the same old state and the same input snapshot. The new tuple commits simultaneously after all required domains and decisions are resolved. A self-loop in an instantaneous operator DAG is inadmissible: every recurrence cycle must cross a declared `DELAY`. An algebraic fixed-point equation is a separate solution declaration needing existence, uniqueness and branch obligations; it is not an execution order.

For w -bit masks, define the retained whole-word operation

$$a = x \wedge A \wedge V, \quad n = a \wedge N, \quad h = \text{popcount}(n \wedge B),$$

$$\text{ASA}(x; V, A, N, B) = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases}$$

The two ordered sets and literal J–K update are

$$\begin{aligned} x_t &= E_X(q_t, d_t), & y_t^A &= \text{ASA}(x_t; V_A, A_A, N_A, B_A), \\ y_t &= \text{ASA}(y_t^A; V_B, A_B, N_B, B_B), \\ J_t &= E_J(q_t, y_t, d_t), & K_t &= E_K(q_t, y_t, d_t), \\ q_{t+1} &= [(J_t \wedge \neg_w q_t) \vee (\neg_w K_t \wedge q_t)] \wedge V_q. \end{aligned}$$

Here $\neg_w q = (2^w - 1) \oplus q$ and each word expression has $\text{Bits}(w)$ type. Both intermediate (a, n, h, y) tuples are observables. Absorption zeroes the whole output of its stage; it neither clears selected bits individually nor silently substitutes a hold transition. The mathematical width w is explicit; the retained executable orbital kernel still uses its specified 32-bit word profile.

Proposition 1.4 (Causal uniqueness and conditional finite periodicity). *For fixed initial states and an input stream, a typed finite instantaneous DAG whose required operations are defined determines at most one next state at each step. If all operations are total on the reachable states, it determines a unique state stream. A fixed autonomous transition on m active Boolean state bits is eventually periodic, with preperiod plus period at most 2^m .*

Proof. Topological induction determines every demanded instantaneous node on the admitted branches from old-state and input values; simultaneous commitment then determines the next tuple. Induction on steps gives the stream. For the finite autonomous case, among $2^m + 1$ states two coincide; determinism makes their subsequent tails coincide. Changing the input, strategy, timestamp, unbounded integer state or physical continuous state does not meet that finite-autonomous hypothesis. $\square$

1.8 Vectors, calculus and orbital expressions

Vector and matrix operators are typed compositions of scalar arithmetic: $u \cdot v = \sum_i u_i v_i$, $\|u\| = \sqrt{u \cdot u}$, and $(AB)_{ij} = \sum_k A_{ik} B_{kj}$. Cross product is restricted to three components. A normalized vector requires nonzero norm; a matrix inverse requires a square matrix and nonzero determinant. Polynomial, Chebyshev, Legendre, finite Fourier and rational expressions expand to declared finite templates. The catalog includes finite sum/product, differentiation, gradient and Jacobian binding, definite integration and an initial-value solution declaration.

$\text{DIFF}(f, x)$ denotes a derivative only where the stated limit exists. $\text{INTEGRAL}(f, a, b)$ denotes an oriented Riemann integral when the finite interval integrand is integrable; an improper integral has a separate limit declaration. $\text{ODE}(F, t_0, s_0, t, I)$ denotes the value at $t \in I$ of the unique continuous state satisfying

$$s(u) = s_0 + \int_{t_0}^u F(v, s(v)) dv \quad (u \in I),$$

with a Riemann-integrable right-hand side along that state. On continuous pieces the solution is differentiable; a coefficient jump need not admit a derivative at the junction. The named continuous FLOW uses this integral-equation meaning. State may be a tuple of position, velocity and other components with different dimensions, including explicitly framed vectors. Each derivative component retains its state's nominal frame and has its state dimension divided by time. Rotating-frame terms must occur in F ; the type annotation does not invent them. A local existence theorem or local regularity condition alone is not a certificate of a global interval. If existence, uniqueness or continuation is unresolved, the node remains a declaration with obligations, rather than an evaluated exact orbit.

A finite RK4 template can be represented exactly as expressions, for example

$$\begin{aligned} k_1 &= F(t, s), & k_2 &= F(t + h/2, s + hk_1/2), \\ k_3 &= F(t + h/2, s + hk_2/2), & k_4 &= F(t + h, s + hk_3), \\ \text{RK4}_h(t, s) &= s + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

Exact evaluation of this finite expression does not make it equal to the continuous ODE solution. Discretization, uncertain initial conditions, unmodelled forces and discontinuous external events remain separate mathematical or physical questions. All force and station expressions must retain their units, frames, absolute time and explicit singular domains. A finite symbolic term is not an error bound on its physical denotation.

1.9 Proof rules and strategy self-improvement

Let Γ contain typed assumptions and proven domain predicates. A permitted rewrite carries the judgement

$$\Gamma \vdash e \simeq e' : \tau, \quad \text{Dom}_\Gamma(e) = \text{Dom}_\Gamma(e'), \quad \llbracket e \rrbracket = \llbracket e' \rrbracket \text{ on that domain.}$$

Domain equality is essential: simplifying x/x to 1 without $x \neq 0$ changes the defined inputs. The proof vocabulary contains assumption, reflexivity, symmetry, transitivity, typed congruence, capture-avoiding substitution, definition expansion and explicit ordered-field/domain lemmas. Branch rules require their predicates; differentiation and integration rewrites additionally require their analytic hypotheses. A rule name or a source digest alone is not a proof of its side conditions.

Structural common-subexpression sharing, exact constant folding and eliminating a pure unused node preserve the denotation of the demanded exports when the typing, scope and domain obligations are satisfied. Reassociation is justified in exact integer/rational arithmetic but is not silently applied to R10 binary64 execution. Eliminating an observed ASA trace, changing a sample read, or changing query cadence is not observationally neutral merely because a final word matches.

For stateful replacement use a relation $\mathcal{R}$ between old and new states. Its proof must relate the initial states and, for every admitted input, preserve definedness, all declared observations and the relation after one synchronous step. Induction then gives equal observable streams. Absolute timestamps and selected sample identities belong to the observations whenever a query schedule is exported; a change in those requires a separately named strategy contract.

A strategy version is an immutable tuple

$$S_v = (\text{rule catalog, priority policy, cost model, proof system ID, resource policy}).$$

It may propose a new expression graph or a new strategy S_{v+1} . Acceptance requires a finite proof object under the already accepted proof system, a preserved input/output contract, and the stated termination/resource argument. The proposal cannot justify itself by replacing its own checker

or dropping an unmet premise. A proof-system change creates a new explicit trust boundary. Strategies and proposal histories may be unbounded, so finite Boolean periodicity does not apply to the combined evolving system.

Proposition 1.5 (Conditional preservation under strategy replacement). *Suppose the accepted proof rules are sound for the declared denotation, each accepted replacement carries a valid domain-preserving proof or stream simulation relation, and proof checking admits only valid derivations. Then any finite chain of accepted replacements preserves its declared observations.*

Proof. A sound proof preserves one replacement's domain and observations. Compose these equalities, or the stepwise stream relations, along the finite chain. Induction gives preservation after every accepted replacement. This establishes a conditional theorem; it supplies neither a proof-checker implementation nor an unconditional claim that proposed derivations are valid. $\square$

Logical strategy improvement can now be stated precisely. Sharing an identical pure subgraph reduces duplicated graph storage; memoizing it avoids a repeated evaluation only when the complete binding, state step and dependency identity are the same. A strictly decreasing natural-valued rewrite measure proves termination of that rewrite phase; a finite proposal budget also terminates a search without proving it finds a better expression. A predicted reduction in operator count is a claim under the declared cost model, not a measured speedup or a guarantee of lower big-integer time, memory use, physical error or global optimality. These are the available reasoning-based improvements in this subversion; execution and machine proof remain future obligations.

2 The exact kernel, physical fields and acquired observations

2.1 A shared state with explicit causal boundaries

The literal operators and one-bit word grammar are fixed in Sections 1 and A. Their application to physics uses the delayed state

$$s_n = (q_n, W_n, \xi_n, \ell_n, c_n; \mathcal{E}_n). \quad (1)$$

Here q_n is the ASA/NA-JK word, W_n an optional WANT word, ξ_n the selected finite physical-model state, ℓ_n consumed-record memory, c_n controller/calibration memory, and $\mathcal{E}_n$ their exact expression roots and immutable dependencies. Unused components are omitted explicitly. An actual plant state $x_{\text{phys}}(t)$ is outside this digital record. Its unknown parameters and disturbances do not become known literals by naming ξ .

Let R_n be a completed immutable measurement record available at control time t_n . Its acquisition interval, source timestamp, availability time, channel validity and calibration binding remain distinct. The following is the selected sample-and-hold chronology, with $h_n = t_{n+1} - t_n > 0$:

$$\begin{aligned} (\widehat{m}_n, e_n) &= C(R_n, c_n), & d_n &= E_D(\widehat{m}_n, e_n, t_n, c_n), \\ (x_n, y_n, J_n, K_n, q_{n+1}^{\text{base}}) &= \mathcal{W}(q_n, d_n), \\ q_{n+1} &= \begin{cases} [(J_n^* \wedge \neg_w q_n) \vee (\neg_w K_n^* \wedge q_n)] \wedge P, & \text{event adapter enabled,} \\ q_{n+1}^{\text{base}}, & \text{event adapter disabled,} \end{cases} \\ (W_{n+1}, \ell_{n+1}, c_{n+1}, u_n) &= \mathcal{A}(W_n, \ell_n, c_n, R_n, q_n, q_{n+1}). \end{aligned} \quad (2)$$

Here P is the present mask. The enabled J_n^*, K_n^* are defined by (5); the command adapter reads this same final q_{n+1} . The model step and its predicted observation are then

$$\begin{aligned} \xi_{n+1} &= \Psi_{\Pi, h_n}(\xi_n, u_n, v_{[t_n, t_{n+1}]}; \theta), \\ \widehat{M}_{n+1} &= O_{\Pi}(\xi_{[t_n, t_{n+1}]}, \theta). \end{aligned} \quad (3)$$

$\mathcal{W}$ is exactly the declared E_X , two ordered independent ASA/NA mask sets, E_J, E_K and synchronous JK composition. All its J/K expressions read the same old q_n ; y_n is the second whole-word output. $C, E_D, \mathcal{A}$ are pure typed templates with complete bodies, including their projection and domain rules. A profile without a WANT transition defines $W_{n+1} = W_n$. Enabling a historical WANT transition requires its complete versioned body; the name W64 alone selects none.

Ψ denotes either the separately specified continuous solution over the interval or a selected finite update. It cannot silently alternate between them. $\mathcal{O}$ needs the selected trajectory or an explicitly defined acquisition reconstruction; two finite endpoint states alone do not define its integral. v comprises declared external loads/inputs; a continuous input needs its finite function definition or an explicit acquisition/model binding. $\hat{M}$ is a model prediction. The actual next sample is separately acquired from x_{phys} and its instrument; it is never substituted by this prediction without an explicitly selected synthetic-input profile. Sensor latency enters the next availability time. A measurement completed after t_n cannot affect the command already applied at t_n . Changing h_n changes the recurrence.

All logical next roots are evaluated from one snapshot and commit together. The physical command and interval schedule form a separate timed interface; an unimplemented mathematical transition is not an actuator. The plant's control authority, command limits and timeout/held-command behavior must be specified by the chosen interface when that interface is used.

2.2 Events, non-values and exact information retention

For the retained Operator I observation, the Boolean event is

$$e_n = X_n \wedge Y_n \wedge C_{t,n} \wedge V_n \wedge \neg S_n \wedge \neg F_n. \quad (4)$$

These are acquired/calibrated threshold, coincidence, validity, saturation and fault facts; they are not inferred from the word masks. The optional event-lane adapter binds one ordered site/session stream, a reserved present-bit mask M_e , and the last consumed identity ℓ_n . For a valid ordered record,

$$\begin{aligned} p_n &= e_n \wedge \text{new}(R_n, \ell_n), & E_n &= \begin{cases} M_e, & p_n, \\ 0, & \neg p_n, \end{cases} \\ J_n^* &= (J_n \wedge \neg_w M_e) \vee E_n, & K_n^* &= (K_n \wedge \neg_w M_e) \vee E_n. \end{aligned} \quad (5)$$

The enabled reserved lane holds for $p_n = 0$ and toggles for $p_n = 1$; other lanes have the base one-step JK result. Later base expressions can read the changed lane, so this is not a global noninterference theorem. Disabling the adapter restores the base J/K expressions. Enabling starts with a recorded baseline without a pulse and consumes only later records. Tracker and word commit atomically. Repeated identical records are a no-new-pulse case; the same identity with changed bytes, calibration or time binding is invalid.

Missing, stale or calibration-unresolved demanded input is an explicit non-value, not a valid false event. Likewise an unknown real comparison is not zero. A selected application's defined failure result is different: the inherited CGK word-first failure can be represented by the single tagged value

$$(q^+, \xi, t, \text{chart history}, \text{failed} = 1), \quad (6)$$

which updates the word while keeping those other fields. A lazy failure branch does not demand its undefined inverse. This remains one atomic transition; it does not authorize a partial commit of a non-value operator. Physical time and external plant evolution continue if a digital transition does not commit.

An exact checkpoint retains sufficient exact operands, roots, source samples and dependencies for the declared state. A rounded rendering is an output projection. Feeding it back instead of

the retained state creates a different recurrence. Source rounding, clipping, modulo and ADC quantization remain explicit functions when selected; the exact word envelope preserves their input and output representations without reversing lost source information.

2.3 W64 is a finite chart/control record

The retained WANT.W64 fields are fixed below. They are distinct from the JK word, spatial UGTS keys and the Operator I measurement packet.

Bits	Field	Interpretation
63	χ	channel/control bit
62:52	ρ	signed 11-bit log-radius code
51:41	ϑ	unsigned phase modulo 2048
40:30	z	signed 11-bit axial code
29:20	Φ	unsigned 10-bit register
19:15; 14:10	depth; phase	two independent 5-bit registers
9:3; 2; 1:0	LUT; κ ; cycle	7-bit index; seam bit; 2-bit selector

Extraction is $(W \gg s) \wedge (2^w - 1)$; signed interpretation is u for $u < 2^{w-1}$ and $u - 2^w$ otherwise. With positive scales, an actual chart is

$$\begin{aligned} r &= r_W 2^{\Delta_\rho \text{signed}_{11}(\rho)}, & \theta &= 2\pi \vartheta / 2048, & z_{\text{phys}} &= \Delta_z \text{signed}_{11}(z), \\ x &= o + r \cos \theta e_1 + r \sin \theta e_2 + z_{\text{phys}} e_3. \end{aligned} \quad (7)$$

The origin o , orthonormal frame (e_1, e_2, e_3) , scales and their calibration are declared inputs. This code has no exact radius-zero point. Decoding the integer tuple is lossless; recovering an arbitrary pre-quantization location from that tuple is not defined. A changed origin or frame needs its full transform. The corresponding log-radius chart uses $\log(r/r_U) = \log(r_W/r_U) + \log(2)\Delta_\rho \text{signed}_{11}(\rho)$. Neither this geometry nor an OTAN2 spelling supplies an optical transfer.

The preserved integer pinion uses $s = (2H_4)v$ and

$$\widehat{A}(v) = \left(\mathcal{R}\left(\frac{106039s_0}{131072}\right), \mathcal{R}\left(\frac{40503s_1}{131072}\right), \mathcal{R}(s_2/2), \mathcal{R}(-s_3/2) \right), \quad \mathcal{R}(x) = \text{sgn}(x) \lfloor |x| + 1/2 \rfloor. \quad (8)$$

Its input is the selected normalized dimensionless coordinate tuple; any subsequent clip, seam or register projection belongs to its full transition template. This exact integer expression is distinct from the algebraic A_ϕ below. Retaining a pre-projection expression does not entitle a control decision to bypass the selected projection.

2.4 Optical fields close through a measured acquisition

The coupled optical chain is

$$U_{\text{det},\lambda} = H_{z,\lambda} \left[A_{\text{ap}} t_\lambda(T, a, \zeta) e^{i\varphi_{\text{lens},\lambda}} U_{\text{in},\lambda} \right]. \quad (9)$$

The electromagnetic chapter supplies Maxwell, material storage/loss and admitted propagation reductions. Here A_{ap} is the physical aperture, a, ζ material states, and φ_{lens} a declared phase field. In a scalar fixed-plane monochromatic profile, normalize U so $|U|^2$ is power per area. In its spectral version use power per area per wavelength; with that corresponding convention define

$$P_\lambda(t) = \int_{\mathcal{D}} |U_{\text{det},\lambda}(x, t)|^2 dA, \quad \mu_k = \int_{t^-}^{t^+} \int_{\Lambda_k} \eta_k(\lambda) \frac{\lambda P_\lambda(t)}{hc} d\lambda dt. \quad (10)$$

In the wavelength integral P_λ is spectral power density; for a discrete monochromatic mode use its power and a finite wavelength sum instead. The spectral power-sum profile requires mutually incoherent components, separately resolved orthogonal frequency channels, or acquisition averaging that removes inter-frequency beat terms. For arbitrary coherent multi-frequency illumination over a finite window, the detector's full field-response equation must retain the relevant cross terms instead. The efficiency $0 \leq \eta_k \leq 1$ assumes at most one primary detection per incident photon and the selected band. μ_k is an ideal expected primary count, not the observed count. Dark counts, gain, saturation, dead time and count statistics belong to the detector model when relevant. A thermal or analogue instrument instead uses its calibrated measurand and acquisition weighting. The actual code is imported with its original bits and exact code-unit map; the measurement chapter supplies its discrepancy, noise and uncertainty terms.

Use one input/field/material energy ledger: polarization storage, transmitted or reflected flux, absorbed energy and heat are the transfers already defined by the selected physical equations. Sensor gain or reporter output does not create a second incident budget. For an ideal binary aperture, $A_{\text{ap}}^2 = A_{\text{ap}}$ and $\|A_{\text{ap}}U\|^2 \leq \|U\|^2$. A general material transmission t is not idempotent; the ASA rule that one hit zeros an entire word is not supplied by this pointwise aperture equation.

2.5 An exact passive target for the algebraic pinion

For dimensionless coordinates, or separately declared power-normalized optical amplitudes, define

$$H_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad \phi = \frac{1 + \sqrt{5}}{2}, \quad A_\phi = \text{diag}(\phi, \phi^{-1}, 1, -1)H_4. \quad (11)$$

$H_4 H_4^T = I$, so the singular values of A_ϕ are $\phi, \phi^{-1}, 1, 1$. Thus A_ϕ is not a passive direct amplitude map for all inputs: $H_4^T e_1$ is a unit input amplified by $\phi > 1$. Instead

$$B = A_\phi / \phi = \text{diag}(1, \phi^{-2}, \phi^{-1}, -\phi^{-1})H_4, \quad B^T B \leq I. \quad (12)$$

For each diagonal entry d_j , pair the mode with an auxiliary mode using

$$O_j = \begin{pmatrix} d_j & \sqrt{1 - d_j^2} \\ -\sqrt{1 - d_j^2} & d_j \end{pmatrix}, \quad O_j^T O_j = I. \quad (13)$$

After H_4 , zero classical auxiliary input amplitude gives B on the selected outputs and carries the remaining power to side ports. This proves an ideal enlarged norm-preserving target with exact algebraic coefficients. A negative d_j requires relative phase; side-port energy is retained. A quantum use also needs its auxiliary states and noise model.

The identity $A_\phi v = \phi(Bv)$ permits an exact symbolic scale factor. It does not restore lost output power or remove measured amplitude uncertainty. An active direct map needs supplied gain and its noise model. A digital photonic encoding of the bits has a different relation between optical power and represented numeric magnitude; it is governed by the next contract.

2.6 Physical realization and minimal specializations

Let $\text{Dec}(p) = s$ relate an admitted physical carrier state to a finite digital state. A physical step $\mathcal{H}_\delta$ under admitted disturbance δ realizes the defined exact step F_Π only if

$$\text{Dec}(\mathcal{H}_\delta(p, \text{Enc}(u))) = F_\Pi(s, u) \quad \text{for every reached admitted } p \text{ with } \text{Dec}(p) = s. \quad (14)$$

The output must again be an admitted representative, or undergo a declared restoration. With the same acquired snapshots and initial decoded state, induction proves agreement for every defined finite prefix within the declared resources. Encoding, carry/borrow, memory, synchronization and atomic commit are obligations of a selected realization; the one-bit transport does not choose a physical carrier or discharge them. For an intensity bit with low level at most a_0 , high level at least a_1 and additive error bounded by ϵ , a threshold satisfying $a_0 + \epsilon < \theta < a_1 - \epsilon$ proves correct decoding. This conditional margin preserves a logical bit, not an unknown analogue measurand to infinite precision.

Only the dependency closure required by an application is imported:

- **Digital.** Omit the plant and detector; retain exact operands, word/expression state, sample ports and declared outputs. A physical hardware claim additionally needs (14).
- **Analogue/mechanical/thermal.** Use the selected finite or continuum physical equations, measured coefficients and boundary ports. A finite basis is an explicit approximation to the continuum, without prescribing a raster. The exact mechanical midpoint identity applies only to its own fixed linear profile; it does not assert conservation for every sampled coupled model.
- **Photonic.** Bind (9) to the material and propagation equations and (10) to the instrument. Choose an analogue transfer or a digital realization explicitly.
- **Orbital/spatial.** Choose $\dot{r} = v$, $\dot{v} = a_{\Pi}(t, r, v)$ with its gravity/forcing, time and frame model. A fixed station observes $z = L(M(t)r - r_g)$, $D = \|z\|$, azimuth $\text{atan2}(z_E, z_N)$ and elevation $\text{atan2}(z_U, \sqrt{z_E^2 + z_N^2})$ on their nonsingular domains. This is simultaneous geometry. A received-signal observation needs its emission time, propagation and clock model. In the inherited orbit seed, JK chooses query cadence and does not enter a_{Π} . Store full time as well as phase: $k - k_0 = Pw + r$, $0 \leq r < P$; (w, r) reconstructs k .
- **Optional attitude.** A unit quaternion Q has kinematic equation $\dot{Q} = \frac{1}{2}Q \otimes [0, \omega]$. A selected sampled correction defines $U = Q + h(\dot{Q}_\omega - \beta e)$ and $Q^+ = U/\sqrt{U^T U}$ for $U^T U > 0$. The residual/gradient e , calibrated rate, gain, frame, bias and missing-sample branches need their complete profile. This proves unit norm after a defined normalization; it does not supply heading, position or filter convergence.

These bindings retain the R6/R10 word lineage, R11 exact grammar, R12 word, optical and event derivations, and R13 physical/measurement contracts. Historical application recipes and results remain versioned dependencies when selected; none is silently promoted into evidence for a new physical device.

3 A minimal physical-law layer for application kernels

A physical law supplies relationships, constrains coupling and supports implementation choices only when its hypotheses are part of the selected model. R14 integrates this law-profile contract, introduced in R13, with the exact representation and state feedback restated in this document.

The unified profile is XOP-UNIFIED-FIELD-R1; its physical-law layer retains the XOP-PHYSICS-R1 definitions. Its mathematical bodies use the existing XOP-R1 operators; it does not allocate new opcodes or claim a new executable backend. A physical specialization imports only the laws and closures needed by its declared interface. A purely digital permutation does not need a continuum electromagnetic model merely because real hardware will eventually execute it.

3.1 Five distinct parts of a physical specialization

Part	What it supplies	Example in the existing model
Balance law	Conserved quantity, flux or production relation	Charge continuity; energy and momentum balance
Constitutive model	Material or component response under stated conditions	Polarization dynamics; thermoelastic free energy
Reduction	A finite or simplified model with its scope	Fixed modal basis; scalar optical field; lumped temperature
Discrete evolution	A selected finite state transition	Existing backward Euler; new energy-balanced midpoint
Observation	How physical state produces recorded evidence	Acquisition window, calibration, quantization and uncertainty

Conservation equations alone do not specify a material. A constitutive law alone does not provide boundary data. Exact arithmetic alone does not select a conservative time update. A recorded count alone does not fix a physical measurand. Keeping these parts explicit allows specialization without hidden changes of meaning.

3.2 Normative profile contract

A selected physical profile declares the finite record

$$\Pi = (\mathcal{T}, \mathcal{X}, \mathcal{U}, \mathcal{Y}, \Theta, \mathcal{B}, \mathcal{C}, \mathcal{A}, \mathcal{N}, \mathcal{O}, \mathcal{E}). \quad (15)$$

$\mathcal{T}$ contains units, frames and time conventions. $\mathcal{X}$ declares state and initial conditions; $\mathcal{U}, \mathcal{Y}$ declare inputs and outputs. Θ contains parameters with source identities and joint uncertainty. $\mathcal{B}$ lists applicable balance laws. $\mathcal{C}$ supplies constitutive closures. $\mathcal{A}$ lists approximation and admissibility hypotheses. $\mathcal{N}$ identifies a finite representation and the chosen update. $\mathcal{O}$ defines acquisition and calibrated observations. $\mathcal{E}$ records the claimed output bounds and evidence status.

Each law entry has a qualified name/version, argument types, equation body, domain, boundary terms, source and a status. Status distinguishes a definition, a derived identity, an adopted physical law, a constitutive assumption and a measured parameter. The same source tag cannot turn one status into another. The editable normative field contract is `formal/PHYSICS_PROFILE.md`. This is a record definition, not a claim that a new parser implements it.

For a finite state model, the mathematical content can be written

$$\dot{x} = f_{\Pi}(t, x, u; \Theta), \quad y = g_{\Pi}(t, x, u; \Theta), \quad B_j(t, x, \dot{x}, u; \Theta) = 0, \quad (16)$$

with declared inequalities and initial/boundary conditions. The B_j are relations to derive or impose consistently, not extra independent equations to append blindly to a fully determined system. Redundant balances are identified; inconsistent initial data or incompatible closures do not define an admitted solution. A finite ODE must meet the existing existence and uniqueness domain. Algebraic constraints need elimination or a separately defined constrained formulation; they are not silently handled by ODE.

3.3 Required connections to the existing seed

Existing quantity	Connection in the unified profile
$t_\lambda, H_{z,\lambda}$	Chosen electromagnetic transfer, material response, boundary data and power normalization
$P_{\text{abs}}, P_{\text{heat}}$	A shared energy ledger distinguishes absorption, stored excitation and dissipated heat
M, C, K and strain	Fixed geometry/basis or declared motion, force/work convention and constitutive response
T, ζ, a	Thermal/material energy stores and rates with the relevant temperature and coupling domains
ASA/NA, JK and actuator command	The word selects an action; a separate actuator equation supplies force, voltage, light or heat
WI measurement and calibration	Acquisition history, physical units, clock/frame binding, uncertainty and measurand equation
Exact operator graph	Captured physical inputs and hypotheses, complete finite operands, explicit unresolved outcomes

For example, a selector $b \in \{0, 1\}$ may choose $K(b) = K_0 + b(K_1 - K_0)$. It does not by itself specify the work required to switch stiffness. For an unchanged coordinate q , that storage change is $q^T(K_1 - K_0)q/2$ and must be attributed to the switching interaction. Signed damping or stiffness remains mathematically expressible; a passive energy theorem applies only when its sign hypotheses hold. An active element needs its actual supply and constitutive model.

3.4 Representation preserves laws only through an explicit map

Let $x = \mathcal{R}(z)$ reconstruct a finite physical model from encoded state z , and let F_h be a chosen digital transition. A continuous-law identity does not automatically give

$$\mathcal{B}_{j,h}(\mathcal{R}(z), \mathcal{R}(F_h(z, u)), u) = 0. \quad (17)$$

Here $\mathcal{B}_{j,h}$ is a separately defined discrete balance, rather than the continuous residual B_j with its arguments silently changed. The discrete relation has to be derived for the selected step. The law may instead provide an error bound or a nonzero residual ledger. Likewise, a finite basis is a declared approximation to a continuum field: exact evaluation of its coefficients and trajectory does not erase omitted modes. The electromagnetic and mechanical chapters state the necessary finite-representation and integral domains.

One-bit word packing preserves the encoded operators, parameters and inputs. It does not turn a finite word mask into an electromagnetic absorber, identify a code bit with a photon, or establish that a material follows a named law. Those are explicit realization mappings. The photonic refinement contract in Section 2 governs a proposed hardware executor.

4 An exact finite step with a derived mechanical energy balance

This addition supplies a specific useful kernel specialization. It removes algorithmic energy loss from the fixed linear model's discrete balance while retaining physical damping. It does not replace the existing CGK or orbital profiles silently and does not assert exact continuous-time trajectories.

4.1 Selected model and units

Choose a finite number of length-valued coordinates q in a fixed declared modal basis, velocity v , and a fixed reference configuration. Let

$$\dot{q} = v, \quad M\dot{v} = \bar{f} - Kq - Cv, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad C = C^T \geq 0. \quad (18)$$

Here the displayed $\bar{f}$ denotes the force selected for the interval's discrete supply; a held-force physical model may use the same constant value. The quantities M, C, K have units kg, kg/s and kg/s² when all q coordinates have length units. Generalized coordinates of mixed dimensions require the inherited tuple/block rules, not a false uniformly dimensioned matrix. The storage is

$$E(q, v) = \frac{1}{2}v^T Mv + \frac{1}{2}q^T Kq, \quad \dot{E} = v^T \bar{f} - v^T Cv. \quad (19)$$

The continuous identity follows by differentiating the two quadratic terms and substituting the model. Its damping term becomes heat or another named output store in a coupled physical model, as derived in the energy chapter.

The parent already defines backward Euler for a more general signed CGK profile. Under fixed passive coefficients its storage change includes

$$-h(v^+)^T Cv^+ - \frac{1}{2}(v^+ - v)^T M(v^+ - v) - \frac{1}{2}(q^+ - q)^T K(q^+ - q) \quad (20)$$

in the unforced case. The last two terms are numerical dissipation from the selected update. Exact arithmetic preserves those terms exactly; it does not remove them. They cannot be reassigned to measured physical heat without a separate modeling decision. R13 therefore names an alternative finite profile.

4.2 PHYS-MIDPOINT-LINEAR-R1

For $h \geq 0$, write $\bar{q} = (q^+ + q)/2$ and $\bar{v} = (v^+ + v)/2$. Define

$$q^+ - q = h\bar{v}, \quad M(v^+ - v) = h(\bar{f} - K\bar{q} - C\bar{v}). \quad (21)$$

These undivided equations also define the $h = 0$ identity step. Eliminate q^+ to obtain the explicit finite solve

$$A_h = M + \frac{h}{2}C + \frac{h^2}{4}K, \quad (22)$$

$$b_h = (M - \frac{h}{2}C - \frac{h^2}{4}K)v - hKq + h\bar{f}, \quad (23)$$

$$v^+ = A_h^{-1}b_h, \quad q^+ = q + \frac{h}{2}(v^+ + v). \quad (24)$$

Proposition 4.1 (Existence and exact field closure). *For the stated symmetric sign conditions and $h \geq 0$, A_h is positive definite. Consequently the step is uniquely defined. If all supplied entries are rational, both output vectors have exact rational entries.*

Proof. For $w \neq 0$, $w^T A_h w = w^T M w + (h/2)w^T C w + (h^2/4)w^T K w > 0$. Thus A_h is invertible. A finite nonsingular linear solve over the rationals uses rational field operations and has rational output. No real root or transcendental decision is required for this selected rational profile. $\square$

Proposition 4.2 (Discrete power balance). *For the same fixed model,*

$$E(q^+, v^+) - E(q, v) = h\bar{v}^T \bar{f} - h\bar{v}^T C\bar{v}. \quad (25)$$

In particular, unforced undamped storage is preserved exactly, while the unforced damped step cannot increase it.

Proof. Symmetry gives $\Delta E = \bar{v}^T M(v^+ - v) + \bar{q}^T K(q^+ - q)$. Substitute (21); the stiffness terms cancel, leaving (25). Nonnegativity follows from $C \geq 0$. $\square$

The supplied work $h\bar{v}^T \bar{f}$ and dissipated energy $h\bar{v}^T C \bar{v}$ are the discrete profile's quantities. They are not automatically the exact work and heat integrals of the continuous trajectory. The theorem is a storage identity, not a proof of arbitrary-step trajectory accuracy or a guarantee about every other invariant.

4.3 Exact seed body and causal integration

The template arguments are

$$(q, v, M, C, K, h, \bar{f}) \mapsto (q^+, v^+, W_{\text{in}}, Q_d), \quad W_{\text{in}} = h\bar{v}^T \bar{f}, \quad Q_d = h\bar{v}^T C \bar{v}. \quad (26)$$

Use existing scalar arithmetic and matrix/vector construction/contraction, then `SOLVE_LINEAR` (opcode 87), to realize (24). The field operations, constants $1/2, 1/4$, and input lineage remain exact expressions. The output ledger is part of the template contract. No proof engine or compiled implementation is supplied here.

All operands read the same old snapshot. A completed event may select the interval's parameters according to the existing causal word policy. If that selection changes stiffness or other storage parameters, first account for the switch at the old state, then apply the fixed-parameter step. The force input and its actuator energy source stay explicit. Both output state roots commit together, and any rounded display remains an export.

This narrow profile makes no claim about time-varying mass, moving bases, nonlinear elasticity, contact, nonlinear reporter kinetics, orbital RK4 or arbitrary signed damping. Those applications retain their own definitions. Nonzero algorithmic residuals in another profile are reported as such; the word “exact” cannot turn them into a conservation proof.

4.4 Law residual as an output certificate

For the selected step, define the exact expression

$$r_E = E(q^+, v^+) - E(q, v) - W_{\text{in}} + Q_d. \quad (27)$$

The preceding algebra proves $r_E = 0$ on the template domain. A future compiler can use that identity as a rewrite or verification obligation without numerically comparing two nearly equal energies. If an implementation returns rounded state, its measured residual concerns that projection and cannot replace the expression-level proof. A residual of zero also does not establish that the physical parameters or observed device trajectory are correct.

5 Momentum, energy and admissible physical coupling

The exact seed needs a rule for composing physical models without creating energy at their interfaces. This chapter adopts established momentum and thermodynamic balances, then restates the finite energy-port contract introduced in R13 for the mechanical and thermal states. These are restrictions on a selected physical model. They do not assign physical meaning to a bit pattern by themselves.

5.1 Momentum uses a realized frame and a declared body

For a closed material body in classical continuum mechanics, let $\rho > 0$ be mass density, v velocity, b body force per unit mass, and σ Cauchy stress. In an inertial Cartesian frame the local balance laws are

$$\dot{\rho} + \rho \nabla \cdot v = 0, \quad \rho \dot{v} = \nabla \cdot \sigma + \rho b, \quad \sigma = \sigma^T. \quad (28)$$

Dots here are material derivatives. Stress symmetry assumes the ordinary continuum without independent body couples or couple stresses. A moving control volume instead requires the corresponding mass and momentum fluxes. These balances and their energy/entropy counterparts are stated in [EN1: Banerjee, Basic Thermoelasticity, sections 1.1–1.3](#).

For a rigid body of fixed mass m , centre of mass r_C , body-to-inertial rotation $R_B \in SO(3)$ and constant body-frame inertia $I_B > 0$, the corresponding finite specialization is

$$m \ddot{r}_C = F_{\text{ext}}, \quad I_B \dot{\omega}_B + \omega_B \times (I_B \omega_B) = \tau_B, \quad (29)$$

$$\dot{R}_B = R_B [\omega_B]_{\times}, \quad [\omega_B]_{\times} a = \omega_B \times a. \quad (30)$$

The torque is about the centre of mass; body components and inertial components must not be mixed. The cross term follows from differentiating angular momentum in rotating axes. See [EN2: MIT Dynamics, lecture 28, Euler's equations](#). For this specialization direct contraction gives

$$\frac{d}{dt} \left(\frac{1}{2} m |\dot{r}_C|^2 + \frac{1}{2} \omega_B^T I_B \omega_B \right) = F_{\text{ext}} \cdot \dot{r}_C + \tau_B \cdot \omega_B, \quad (31)$$

because $\omega_B \cdot [\omega_B \times (I_B \omega_B)] = 0$. An inertia estimate, applied torque and orientation measurement are different inputs; a Madgwick attitude estimate alone supplies no torque law.

The XOP frame annotations preserve coordinate identity. Their physical realization also requires origins, alignment and time convention. Transforming to accelerating or rotating coordinates changes the acceleration equation; renaming an inertial force law as a station-frame law is insufficient. Likewise a superposed rigid rotation must not generate elastic strain energy. For a finite-deformation hyperelastic model with deformation gradient $F = \nabla_X x$, $\det F > 0$, the selected objective free energy obeys

$$\Psi(QF, T) = \Psi(F, T), \quad Q \in SO(3), \quad (QF)^T (QF) = F^T F. \quad (32)$$

Material directions, if present, remain explicit reference data; objectivity does not imply isotropy. This condition and the two-point transformation of F are discussed in [EN3: Bouchbinder, Non-Equilibrium Continuum Physics, section 4](#). The existing $\epsilon = \text{sym } \nabla u$ model remains a small-gradient approximation; equation (32) does not extend it to arbitrary finite rotations. A finite-rotation specialization must supply its own objective constitutive equation and compatible finite state.

5.2 A finite port contract with an explicit power proof

Use the finite state x , differentiable storage $H(x)$, and matrices $\mathcal{J}(x) = -\mathcal{J}(x)^T$, $\mathcal{R}(x) = \mathcal{R}(x)^T \geq 0$ and $\mathcal{G}(x)$. Define

$$\dot{x} = (\mathcal{J} - \mathcal{R}) \nabla H + \mathcal{G}u, \quad y = \mathcal{G}^T \nabla H. \quad (33)$$

This is the standard finite port-Hamiltonian form; $u^T y$ is power supplied at the declared boundary. See [EN4: van der Schaft, Port-Hamiltonian nonlinear systems, equations 5–6](#). The symbol $\mathcal{J}$ is unrelated to the JK input word J . Each contraction must be dimensionally compatible. If components of x have different units, their derivative and matrix entries have corresponding component units; they are stored as typed tuples and explicit contractions, not forced into a homogeneous vector type.

Proposition 5.1 (Finite power balance and composition). *For a differentiable solution of (33),*

$$\dot{H} = u^T y - \mathcal{D}, \quad \mathcal{D} = (\nabla H)^T \mathcal{R} \nabla H \geq 0. \quad (34)$$

For two such systems whose ports admit a dimensionally compatible map L , connect $u_1 = L y_2$, $u_2 = -L^T y_1$. Their internal power cancels. Their combined storage is $H_1 + H_2$ and its rate is minus their dissipation plus any remaining external port power.

Proof. Apply the chain rule and use $z^T \mathcal{J} z = 0$ for a skew matrix. The two connected powers sum to $y_2^T L^T y_1 - y_1^T L y_2 = 0$. The entries of L carry any required unit and frame conversion; a force is not silently identified with a velocity. $\square$

If H has a lower bound, integration gives the usual passivity inequality $H(t) - H(0) \leq \int_0^t u^T y \, ds$. This limits extraction relative to storage and supply; it does not by itself prove asymptotic stability, accurate tracking, or a sufficient actuator. Algebraic port connections must also yield a well-defined finite evolution.

Physical port pairs include force/velocity, torque/angular velocity and voltage/current. In a thermal port the energy pairing is absolute temperature times entropy flow, $T \dot{S}_{\text{transfer}} = \dot{Q}$ at that port. Pairing temperature directly with heat power and calling the product power would have the wrong units. Entropy production inside the body is distinct from entropy transported through a port.

5.3 The existing modal mechanics fits this contract

R12 already declares a finite displacement expansion $u_m(X, t) = \sum_i q_i(t) \phi_i(X)$, mass and stiffness matrices from its weak form, and support conditions. For a fixed basis, fixed material parameters, isothermal or explicitly uncoupled mechanics, take

$$M \ddot{q} + D \dot{q} + K q = B u, \quad M = M^T > 0, \quad K = K^T \geq 0, \quad D = D^T \geq 0. \quad (35)$$

Let $p = M \dot{q}$ and

$$H_m(q, p) = \frac{1}{2} p^T M^{-1} p + \frac{1}{2} q^T K q, \quad \mathcal{J} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad \mathcal{R} = \begin{bmatrix} 0 & 0 \\ 0 & D \end{bmatrix}, \quad \mathcal{G} = \begin{bmatrix} 0 \\ B \end{bmatrix}. \quad (36)$$

Then (33) gives exactly (35), with

$$y = B^T \dot{q}, \quad \dot{H}_m = u^T B^T \dot{q} - \dot{q}^T D \dot{q}. \quad (37)$$

The PSD stiffness admits free modes; asymptotic displacement stability does not follow when rigid or undamped modes remain. If M depends on configuration, the derivative of its kinetic energy is part of the dynamics; merely replacing constant M by $M(q)$ in (35) generally misses terms.

Finite modal reduction preserves this displayed energy identity when its coefficient assumptions hold. It does not prove the basis resolves every physical mode. Spatial integrals still require the charts, measures and integrability obligations already specified by R12. A continuum PDE printed above is an origin for the model; it is not an implemented infinite-dimensional XOP operator.

An actual ASA/NA–JK specialization provides an explicit decode from its completed old state to a command, then a transduction law to the physical effort $u(t)$. A voltage command cannot be substituted as a force without that law. The resulting work is

$$W_n = \int_{t_n}^{t_{n+1}} u(t)^T y(t) \, dt. \quad (38)$$

The command may be held over the interval; the velocity generally is not. A bit selecting a positive effort can supply energy through the actuator's power source. Neither bit packing nor a delayed update makes that port passive.

5.4 Convert dissipated work to heat exactly once

For the uncoupled mechanical storage above let $P_d = \dot{q}^T D \dot{q} \geq 0$. Suppose all this dissipated mechanical power thermalizes in the modeled assembly, and fixed or state-declared fractions $w_i \geq 0$, $\sum_i w_i = 1$, distribute it among thermal nodes. Write

$$\dot{H}_m = P_{\text{mech,in}} - P_d, \quad (39)$$

$$C_i \dot{T}_i = P_{i,\text{other}} + w_i P_d + \sum_{j \neq i} G_{ij}(T_j - T_i) + G_{i0}(T_a - T_i), \quad (40)$$

where $C_i > 0$, $G_{ij} = G_{ji} \geq 0$ and $G_{i0} \geq 0$. The stores $U_i = C_i(T_i - T_{i,\text{ref}})$ are an uncoupled constant-capacity model. Their sum with H_m obeys the directly derived identity

$$\frac{d}{dt}(H_m + \sum_i U_i) = P_{\text{mech,in}} + \sum_i P_{i,\text{other}} + \sum_i G_{i0}(T_a - T_i). \quad (41)$$

Internal conduction cancels in pairs, as does mechanical-to-thermal power. Thus P_d is not added again as an external input. If only a fraction thermalizes locally, the remaining energy needs its destination, such as outgoing sound or a separate store. A resistor's VI and an optical absorber's thermalized energy are handled with the same nonduplicating boundary ledger. The distinction between work conversion and entropy creation is treated in [EN5: MIT, Work-to-heat Transduction](#).

5.5 Entropy supplies a second, independent condition

Conserving total energy does not make every proposed heat exchange physical. Restrict the node model to local equilibrium with $T_i > 0$, and define

$$S_i - S_{i,\text{ref}} = C_i \ln(T_i/T_{i,\text{ref}}), \quad T_{i,\text{ref}} > 0, \quad \dot{S}_i = C_i \dot{T}_i/T_i. \quad (42)$$

For one reciprocal conductive link the derived total entropy rate is

$$\dot{S}_{ij} = \frac{G_{ij}(T_j - T_i)}{T_i} + \frac{G_{ij}(T_i - T_j)}{T_j} = \frac{G_{ij}(T_i - T_j)^2}{T_i T_j} \geq 0. \quad (43)$$

A link to an ideal bath at $T_a > 0$ has the same expression after including the bath's entropy change. Local thermalization of $w_i P_d$ contributes $w_i P_d/T_i \geq 0$. Arbitrary external heat, light, matter and work ports must carry their own energy and entropy accounting; an incident optical field's entropy flux is not assigned the thermal formula P_{optical}/T_i merely because its energy is later absorbed.

In local continuum heat conduction, with heat flux $q_h = -\kappa \nabla T$ and positive-semidefinite symmetric conductivity κ , the analogous production density is

$$\xi_{\text{cond}} = q_h \cdot \nabla(1/T) = \frac{(\nabla T)^T \kappa \nabla T}{T^2} \geq 0. \quad (44)$$

Fourier's constitutive law is a local-equilibrium model, not a universal description of ballistic transport or ultrafast material response. Entropy flux and production are distinguished in [EN6: MIT, Entropy Production and Nonlinearity](#) and [EN7: MIT Advanced Thermodynamics, lecture 21, heat conduction](#).

5.6 Thermoelastic expansion requires reciprocal heat coupling

R12's thermal strain is useful, but a coupled model cannot independently choose an elastic energy, expansion force and unrelated heat capacity. The following small-gradient, local-equilibrium

specialization derives them from one Helmholtz free-energy density $\psi(\epsilon, T)$ per reference volume. All densities below use that same small-strain reference convention. Define

$$s = -\psi_T, \quad e = \psi + Ts = \psi - T\psi_T, \quad \sigma = \psi_\epsilon + \sigma_d, \quad c_\epsilon = -T\psi_{TT} > 0. \quad (45)$$

Here s is entropy density, e internal energy density, σ_d dissipative stress, and c_ϵ volumetric heat capacity at fixed strain. These thermodynamic conjugacies follow the free-energy formulation in [EN1, sections 1.3.3–1.3.4](#); the following compatibility calculation is the R13 specialization.

Combining the local balances $\dot{e} = \sigma : \dot{\epsilon} - \nabla \cdot q_h + r$ and $\dot{s} + \nabla \cdot (q_h/T) - r/T = \xi \geq 0$ gives

$$T\xi = \sigma : \dot{\epsilon} - \dot{\psi} - s\dot{T} - q_h \cdot \nabla T/T, \quad (46)$$

$$= \sigma_d : \dot{\epsilon} - q_h \cdot \nabla T/T \geq 0, \quad (47)$$

$$c_\epsilon \dot{T} = T\psi_{T\epsilon} : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (48)$$

The last equation follows by differentiating $e = \psi - T\psi_T$ and canceling $\psi_\epsilon : \dot{\epsilon}$. Here r is heat supply after any declared local conversion, rather than a second copy of mechanical dissipation. With $\sigma_d = \mathbb{V} : \dot{\epsilon}$, symmetric $\mathbb{V} \geq 0$ on symmetric strains, and Fourier conductivity as above, both displayed dissipative contributions are nonnegative. The reversible term $T\psi_{T\epsilon} : \dot{\epsilon}$ need not be positive: it exchanges thermal and mechanical storage without being dissipation.

A concrete minimal profile uses constant elastic tensor $\mathbb{C}$ positive on admissible strains, constant symmetric thermal-stress tensor β , constant $c_v > 0$ and $T_0 > 0$:

$$\psi(\epsilon, T) = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon - (T - T_0)\beta : \epsilon + c_v [T - T_0 - T \ln(T/T_0)], \quad (49)$$

$$\sigma = \mathbb{C} : \epsilon - (T - T_0)\beta + \sigma_d, \quad (50)$$

$$s = \beta : \epsilon + c_v \ln(T/T_0), \quad (51)$$

$$e = \frac{1}{2}\epsilon : \mathbb{C} : \epsilon + T_0\beta : \epsilon + c_v(T - T_0), \quad c_\epsilon = c_v, \quad (52)$$

$$c_v \dot{T} = -T\beta : \dot{\epsilon} + \sigma_d : \dot{\epsilon} - \nabla \cdot q_h + r. \quad (53)$$

Choosing $\beta = \mathbb{C} : (\alpha_T I)$ reproduces R12's constant-coefficient thermal-expansion stress. The linear term in e is required by the selected entropy convention and coupling; replacing e by an unrelated “elastic plus $c_v T$ ” formula breaks the same first-law calculation. The profile applies on a declared strain and positive-temperature range; it is not a fitted constitutive law for an unspecified specimen.

Additional reporter, damage or chemical state z changes this closure to $\psi(\epsilon, T, z)$ and adds $-\psi_z \cdot \dot{z}$ to $T\xi$. An interval-preserving reporter update from R12 proves its range, not this new thermodynamic inequality. Either specify the driving energy/rates that meet it, or keep the reporter as an empirical readout state with its energy exchange separately declared. No new chemical model is assumed here.

5.7 Switching and exact steps keep their own energy obligations

If a physical mode a changes the storage function, continuous parameter motion has the additional power term $H_a \dot{a}$ in the chain rule. At an idealized discrete switch, use one consistent reference for the stores and record

$$\Delta H_{\text{switch}} = H_{a^+}(x^+) - H_{a^-}(x^-) = W_{\text{switch,in}} - E_{\text{switch,out}}. \quad (54)$$

An ideal routing switch can have zero storage jump. A change of stiffness at fixed q, p and fixed M instead contributes $\frac{1}{2}q^T(K^+ - K^-)q$; a state reset or impulsive motion has its own mechanical balance. Labeling the selector as ASA, NA, JK or a thermal bit does not remove these terms. Passive-mode energy inequalities cannot simply be concatenated across unaccounted jumps.

Exact evaluation preserves the expression of a selected time-step map. It does not automatically preserve (34) between its endpoints. Section 4 supplies a finite midpoint specialization with its own derived step energy identity for fixed linear mechanical coefficients. The continuous coupled thermoelastic equations here retain their separate finite-reduction, existence and time-discretization obligations.

5.8 What is added to the kernel contract

No opcode is needed for these laws. The unified profile binds energy and entropy functions, their derivatives, domain conditions, port pairings and selected finite evolution as versioned templates using existing exact arithmetic, LN, finite integrals and finite ODE declarations. States of unlike units use tuples; matrix contractions expand componentwise. A valid specialization can now present its power identity, heat-transfer cancellation and nonnegative entropy production as explicit mathematical obligations. A false sign, unpowered actuator, duplicated heat term or incompatible thermal expansion can be identified before a kernel is run. This is the useful added foundation. Specimen parameters, model adequacy, runtime evaluation and physical performance remain separate evidence.

6 Electromagnetic closure of the optical interface

R12 defines $U_{\text{det},\lambda} = H_{z,\lambda}[A t_{\lambda} e^{i\phi_{\text{lens},\lambda}} U_{\text{in},\lambda}]$. This chapter supplies a physical route to its transfer, loss and material-state arguments. Maxwell's equations are the adopted classical field laws; a material law, geometry and boundary data select a particular device. The added templates fit the existing exact expression and finite-state interfaces.

6.1 Fields, charge and the data that close their evolution

Use SI quantities in one declared spatial frame and time coordinate. For a stationary macroscopic medium, with free charge ρ_f and free current $\mathbf{J}_f$, adopt

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho_f, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}, & \nabla \times \mathbf{H} &= \mathbf{J}_f + \partial_t \mathbf{D}. \end{aligned} \quad (55)$$

The dimensions are $[\mathbf{E}] = \text{V/m}$, $[\mathbf{H}] = \text{A/m}$, $[\mathbf{D}] = \text{C/m}^2$, $[\mathbf{B}] = \text{T}$, $[\rho_f] = \text{C/m}^3$ and $[\mathbf{J}_f] = \text{A/m}^2$. The divergence of the last equation gives

$$\partial_t \rho_f + \nabla \cdot \mathbf{J}_f = 0. \quad (56)$$

Thus supplied charge and current histories cannot be chosen independently. The electric divergence constraint propagates when it holds initially and the supplied charge obeys continuity; the magnetic constraint propagates because the divergence of a curl vanishes. These statements require the corresponding classical derivatives or an explicitly declared weak solution. The laws and charge-conservation argument are documented in the [Caltech Feynman Lectures, II.18 \[EM1\]](#).

For a stationary interface, let $\mathbf{n}$ point from medium 1 to medium 2. With surface free charge ρ_s , surface free current $\mathbf{K}_s$, and no magnetic surface sources, the interface conditions are

$$\begin{aligned} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= \rho_s, & \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) &= 0, \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) &= 0, & \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) &= \mathbf{K}_s. \end{aligned} \quad (57)$$

They follow by shrinking Maxwell's integral pillboxes and loops across the interface. A moving boundary needs the corresponding moving-medium model; these stationary formulas cannot silently serve as that model. [MIT Electromagnetics and Applications, Section 2.6 \[EM2\]](#).

A device profile binds its material regions, interface geometry, constitutive relations, sources, compatible initial fields and initial material memory. It selects the needed independent boundary or incoming-wave data, including an outgoing-wave condition for an open problem. It does not prescribe every field component independently on every boundary. Existence and uniqueness belong to this completed problem; the four field equations alone do not resolve a resonant, contradictory or underspecified setup.

6.2 Power balance before optical loss is called heat

The vector identity for $\nabla \cdot (\mathbf{E} \times \mathbf{H})$ and equation (55) give the constitutive-independent identity

$$\nabla \cdot \mathbf{S} + \mathbf{E} \cdot \partial_t \mathbf{D} + \mathbf{H} \cdot \partial_t \mathbf{B} = -\mathbf{E} \cdot \mathbf{J}_f, \quad \mathbf{S} = \mathbf{E} \times \mathbf{H}. \quad (58)$$

For instantaneous, time-independent real symmetric positive-definite tensors ϵ, μ , with $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$, set

$$u_{\text{em}} = \frac{1}{2} \mathbf{E} \cdot \epsilon \mathbf{E} + \frac{1}{2} \mathbf{H} \cdot \mu \mathbf{H},$$

$$\partial_t u_{\text{em}} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_f. \quad (59)$$

The quadratic energy follows by differentiating each form; time independence and symmetry are used at that exact step. This is Poynting's energy balance, with fields doing work on matter through $\mathbf{E} \cdot \mathbf{J}_f$. [Caltech Feynman Lectures, II.27 \[EM3\]](#).

Writing $\mathbf{J}_f = \sigma \mathbf{E} + \mathbf{J}_{\text{src}}$, where the symmetric part of σ is positive semidefinite, identifies a nonnegative Ohmic sink $q_{\text{Ohm}} = \mathbf{E} \cdot \sigma \mathbf{E}$. The impressed source term $-\mathbf{E} \cdot \mathbf{J}_{\text{src}}$ may supply field energy. In contrast, arbitrary $\mathbf{E} \cdot \mathbf{J}_f$ need not be positive or entirely heat: work can enter mechanical or other material stores. A dispersive dielectric also cannot use $\frac{1}{2} \mathbf{E} \cdot \mathbf{D}$ as its complete instantaneous energy without a justified material-state model.

6.3 A finite dispersive material law with an energy proof

An optional, explicit local linear closure uses finitely many polarization oscillators. In a fixed material region let $\epsilon_b, \mu > 0$ be scalar, time-independent background coefficients and set

$$\mathbf{D} = \epsilon_b \mathbf{E} + \sum_{j=1}^m \mathbf{P}_j, \quad \mathbf{B} = \mu \mathbf{H},$$

$$\dot{\mathbf{P}}_j = \mathbf{V}_j, \quad \dot{\mathbf{V}}_j = f_j \mathbf{E} - \gamma_j \mathbf{V}_j - \omega_j^2 \mathbf{P}_j, \quad (60)$$

where m is finite, $f_j > 0$, $\gamma_j \geq 0$ and $\omega_j > 0$. Here $[f_j] = [\epsilon_b]/s^2$, $[\gamma_j] = s^{-1}$ and $[\omega_j] = s^{-1}$. Each oscillator has supplied $\mathbf{P}_j(t_0), \mathbf{V}_j(t_0)$. This is a selected Lorentz material model, not a universal constitutive law. Its optical origin is described in [MIT Fundamentals of Photonics, Section 2.1.4 \[EM4\]](#).

Proposition 6.1 (Explicit storage and dissipation). *For the held coefficients in equation (60), define*

$$\begin{aligned} u_j &= \frac{|\mathbf{V}_j|^2 + \omega_j^2 |\mathbf{P}_j|^2}{2f_j}, \\ u_* &= \frac{1}{2}\epsilon_b |\mathbf{E}|^2 + \frac{1}{2}\mu |\mathbf{H}|^2 + \sum_j u_j, \\ q_{\text{diss}} &= \mathbf{E} \cdot \sigma \mathbf{E} + \sum_j \frac{\gamma_j}{f_j} |\mathbf{V}_j|^2 \geq 0. \end{aligned} \quad (61)$$

Then $\partial_t u_* + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}_{\text{src}} - q_{\text{diss}}$.

Proof. Multiplying the oscillator equation by $\mathbf{V}_j/f_j$ gives $\dot{u}_j = \mathbf{E} \cdot \mathbf{V}_j - \gamma_j |\mathbf{V}_j|^2/f_j$. In equation (58), the polarization part of $\mathbf{E} \cdot \dot{\mathbf{D}}$ is $\sum_j \mathbf{E} \cdot \mathbf{V}_j$. Adding the oscillator balances cancels these exchanges exactly. Positivity follows from the coefficient and conductivity hypotheses. $\square$

This supplies a consistent source for R12's thermal ledger. Assign $P_{\text{heat,opt}} = \int_{\Omega} q_{\text{diss}} dV$ only when these dissipative channels thermalize in the modeled thermal store on the declared interval. If excitation, emission or another store participates, partition the sink and retain that store's evolution. Counting the same absorbed power independently as full heat and full chemical excitation would break the balance.

With the phasor convention $\mathbf{E}(t) = \text{Re}(\widehat{\mathbf{E}}e^{-i\omega t})$, $\omega > 0$, the scalar isotropic, steady harmonic closure is

$$\begin{aligned} \epsilon_{\text{eff}}(\omega) &= \epsilon_b + \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} + \frac{i\sigma}{\omega}, \\ \langle q_{\text{diss}} \rangle &= \frac{\omega}{2} \text{Im}(\epsilon_{\text{eff}}) |\widehat{\mathbf{E}}|^2. \end{aligned} \quad (62)$$

All denominators must be nonzero, and the selected steady response must exist; an undamped resonance does not satisfy these assumptions. Conductivity is included once, inside ϵ_{eff} in this frequency convention. The time-domain state in equation (60) remains the definition when a steady response is inappropriate.

Temperature, hysteresis and reporter state may parameterize a family of these material profiles. Holding them fixed during an optical subproblem is an explicit time-scale approximation. If coefficients change during propagation, their derivatives change stored energy: for example, $\partial_t(\epsilon_b |\mathbf{E}|^2/2)$ includes $\dot{\epsilon}_b |\mathbf{E}|^2/2$. The corresponding modulation work and material exchange must be included. Replacing a coefficient at each tick without that accounting is not the proved coupled energy law.

6.4 How Maxwell closes the existing propagation operator

In a homogeneous, source-free, lossless region with constant scalar $\epsilon, \mu > 0$, the divergence constraint gives $\nabla \cdot \mathbf{E} = 0$. Taking a curl and using $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ yields

$$\begin{aligned} \nabla^2 \mathbf{E} - \mu\epsilon \partial_t^2 \mathbf{E} &= 0, \\ (\nabla^2 + k^2) \widehat{\mathbf{E}} &= 0, \quad k = \omega\sqrt{\mu\epsilon}. \end{aligned} \quad (63)$$

For a linear dispersive homogeneous medium, use the corresponding frequency equation with its declared complex constitutive coefficients and radiation branch. A scalar component can stand for the optical field only within an appropriate polarization/geometry reduction. Spatially varying tensors, strong polarization conversion or near-field boundary interactions do not automatically reduce to the same scalar equation. [MIT Optics, Lecture 9 \[EM5\]](#).

For the scalar, forward, homogeneous lossless case write $U = e^{ikz}\psi$. Substitution gives

$$\partial_z^2 \psi + 2ik\partial_z \psi + \nabla_\perp^2 \psi = 0. \quad (64)$$

The paraxial approximation drops $\partial_z^2 \psi$ when it is negligible relative to the retained terms on the admitted field class. For $z > 0$, one resulting Fresnel transfer, with $\lambda_m = 2\pi/k$, is

$$(H_z U_0)(x, y) = \frac{e^{ikz}}{i\lambda_m z} \iint U_0(x', y') \exp\left(\frac{ik[(x - x')^2 + (y - y')^2]}{2z}\right) dx' dy'. \quad (65)$$

This states the normalization missing from a mere angular-shape transform. For R12's finite aperture, the integral is over its bounded support; its exact real-pair lift needs the section integrability conditions already specified there. A full-plane operator has its separately declared function-space domain. Fraunhofer additionally neglects the aperture-plane quadratic phase on the stated field/support scale. These approximations need application error allowances, not an assertion that exact encoding makes them exact Maxwell solutions. [MIT Optics, Lecture 16 \[EM6\]](#).

The same boundary-value problem fixes t_λ and reflected/outgoing channels from material parameters and geometry. A thin local multiplier is an adopted reduction; a general scattering device may require a matrix or integral transfer. The aperture A remains a geometric predicate. It does not specify the polarization currents or electromagnetic boundary response of the physical opaque material by itself.

6.5 Passivity at the same physical interface

For a fixed volume, let $\mathcal{U} = \int_\Omega u_* dV$. Separate its boundary flux into admitted incoming and outgoing power. With no impressed source, integration of the derived energy identity gives

$$E_{\text{out}}[t_0, t_1] \leq E_{\text{in}}[t_0, t_1] + \mathcal{U}(t_0) - \mathcal{U}(t_1). \quad (66)$$

Stored energy can temporarily power an output. In steady harmonic operation, the average storage change vanishes; for independent positive-flux modal coordinates this gives precisely R12's condition

$$S^\dagger W_{\text{out}} S \leq W_{\text{in}}. \quad (67)$$

Both metrics refer to the same physical normalization and selected ports. Unobserved reflected, radiated or coupled-away power is not automatically absorption. Only with all outgoing channels accounted for, no supplied gain and no net storage accumulation does $\mathcal{R} + \mathcal{T} + \mathcal{A} = 1$ identify the remaining fraction as material absorption. This derivation supplies the physical premise for the R12 scaled passive pinion construction; it does not change its algebra or demonstrate a manufactured device.

One optional mechanical coupling is already useful without a general electromagnetic stress model. For a stationary planar target in vacuum, normal incidence, forward transmission and specular backward reflection, the steady force along the incident beam is

$$F_{\parallel} = \frac{P_{\text{in}}}{c} (1 + \mathcal{R} - \mathcal{T}) = \frac{P_{\text{in}}}{c} (\mathcal{A} + 2\mathcal{R}). \quad (68)$$

It follows by subtracting outgoing from incoming momentum flux. It can enter the declared mechanical load. Oblique/diffuse scattering, moving media, directional re-emission or additional forces require their corresponding momentum balance; the scalar formula does not cover them. [Caltech Feynman Lectures, I.34.9 \[EM7\]](#).

6.6 Finite exact seed representation

Complex coefficients, amplitudes and matrices use R12's ordered real-pair templates. For a declared finite modal problem $Az = b$, with $A = X + iY$, $z = p + iq$ and $b = r + is$, the literal real system is

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} r \\ s \end{pmatrix}. \quad (69)$$

An inverse requires a proved nonzero determinant; an unresolved singularity or branch remains an explicit non-value outcome. The coefficients bind the chosen basis, geometry, boundary projection, material law and frequency. They may be retained exact expressions or supplied calibrated literals with uncertainty. For instance, select a bounded fixed cavity with a perfect electric conductor boundary and real admissible basis functions ψ_j whose tangential traces vanish. The expansion $\widehat{\mathbf{E}}_N = \sum_{j=1}^N z_j \psi_j$ and weak projection of the frequency curl-curl equation give

$$\begin{aligned} A_{ab} &= \int_{\Omega} [(\nabla \times \psi_a) \cdot \mu^{-1}(\nabla \times \psi_b) - \omega^2 \psi_a \cdot \epsilon_{\text{eff}} \psi_b] dV, \\ b_a &= i\omega \int_{\Omega} \psi_a \cdot \widehat{\mathbf{J}}_{\text{src}} dV. \end{aligned} \quad (70)$$

The boundary term vanishes under this particular trace condition. The finite space, source projection, divergence constraints and coefficient integral domains must be specified. Other boundaries require their own terms. These volume-integral definitions fit existing exact operators only when their actual integrands and bounded coordinate domains meet that operator contract; the weak notation alone is not an executable integral encoding.

The finite polarization coordinates use the first-order equations in (60). Together with a declared finite field reduction, they form an ordinary finite state evolution only after building algebraic constraints into the basis or explicitly eliminating them to obtain a nonsingular finite evolution. The fixed representation, selected elimination branch and regular domain remain explicit. XOP can denote the resulting ODE under its existence/domain contract or an explicitly selected finite update. Remaining algebraic constraints need their own defined constrained formulation. It does not acquire an arbitrary continuum PDE solver. Basis truncation, omitted modes and an approximate constitutive fit remain model errors even when every retained operation and packed bit is exact. The physical profile therefore exports the field transfer, dissipative/other energy channels and optional force from one common model, rather than fitting unrelated outputs that happen to share a seed identifier.

7 One coupled field state and one physical energy

The physical unification supplied here is an explicit composition of electromagnetic, polarization, mechanical and thermal state. Its coupling terms come from one energy function and its dissipative channels have named destinations. Maxwell's equations, material constitutive assumptions and thermodynamic balances keep their established meanings. No additional force law is inferred from a shared notation.

R13 already supplies separate field and mechanical stores, a Lorentz material law and a thermoe-lastic free energy. The missing connection addressed here is a common evolution when those stores depend on the same material or mechanical state. Independently changing optical coefficients, adding a force and depositing absorbed power as heat need not conserve their combined energy. The following construction makes those dependencies part of the formalization.

7.1 The common continuous model

Let z be a finite tuple of nonthermal state coordinates and let $s = (s_1, \dots, s_n)$ contain local-equilibrium entropy coordinates, in J/K. Choose a twice continuously differentiable total energy $\mathcal{H}(z, s)$, in joules, and define

$$e_z = \partial_z \mathcal{H}, \quad T_i = \partial_{s_i} \mathcal{H} > 0, \quad S_{\text{tot}} = \sum_i s_i. \quad (71)$$

These derivatives hold all other state coordinates fixed. Distinct physical units are retained componentwise; a matrix below represents the corresponding typed contractions. The temperature is the derivative of the *complete* energy, including its constitutive coupling, rather than an unrelated lookup. The use of thermodynamic potentials and their natural-variable derivatives is established in [UC1: Alberty et al., Use of Legendre Transforms in Chemical Thermodynamics](#).

Choose $\mathcal{J}^T = -\mathcal{J}$, dissipative maps $\mathcal{R}_\ell = \mathcal{R}_\ell^T \geq 0$, an input map $\mathcal{G}$ and nonnegative allocation fractions $\alpha_{i\ell}$ with $\sum_i \alpha_{i\ell} = 1$. Let $k_{ij} = k_{ji} \geq 0$ be thermal conductances in W/K, with $k_{ii} = 0$, and Q_i be externally supplied heat rates in watts. Define the versioned coupled model

$$\begin{aligned} \dot{z} &= \mathcal{J}e_z - \sum_\ell \mathcal{R}_\ell e_z + \mathcal{G}u, & y &= \mathcal{G}^T e_z, \\ D_\ell &= e_z^T \mathcal{R}_\ell e_z \geq 0, & \dot{s}_i &= \frac{\sum_\ell \alpha_{i\ell} D_\ell + \sum_j k_{ij}(T_j - T_i) + Q_i}{T_i}. \end{aligned} \quad (72)$$

The input/output pairing $u^T y$ is supplied nonthermal power. Each D_ℓ is power removed from the nonthermal coordinates and thermalized in the declared stores on the modeled time scale. A channel that stores excitation, emits radiation or exports power must retain that state or output instead of being wholly allocated as heat. The heat entropy term Q_i/T_i uses the receiving node's local-equilibrium temperature. An external reservoir at a different temperature needs its own entropy flux and interface production; coherent incident optical power is not assigned a heat entropy P/T_i .

This is a selected finite thermodynamic composition. Energy-preserving, entropy-producing coupling is part of established irreversible port-Hamiltonian modeling; see [UC2: Philipp et al., section 2.1](#). The explicit identities below follow directly from the equations here; no control, stability or convergence result from that source is imported.

Proposition 7.1 (Combined first and second laws). *For a differentiable solution in the stated positive-temperature domain,*

$$\dot{\mathcal{H}} = u^T y + \sum_i Q_i, \quad (73)$$

$$\dot{S}_{\text{tot}} - \sum_i \frac{Q_i}{T_i} = \sum_{i,\ell} \frac{\alpha_{i\ell} D_\ell}{T_i} + \sum_{i < j} \frac{k_{ij}(T_i - T_j)^2}{T_i T_j} \geq 0. \quad (74)$$

Proof. The chain rule gives $\dot{\mathcal{H}} = e_z^T \dot{z} + \sum_i T_i \dot{s}_i$. The skew contraction vanishes. Each $-D_\ell$ is canceled by $\sum_i \alpha_{i\ell} D_\ell$; the two heat rates on every reciprocal conduction link cancel. This proves the energy identity. Summing the entropy equations pairs a link's terms as $k_{ij}[(T_j - T_i)/T_i + (T_i - T_j)/T_j] = k_{ij}(T_i - T_j)^2/(T_i T_j)$. All remaining production terms are nonnegative. $\square$

Thus an isolated selected model conserves its total energy while dissipation increases entropy. Energy has not vanished when mechanical or field amplitude decays: it has entered the same model's thermal state. Positivity hypotheses are constitutive assumptions, not conclusions about every proposed device.

The maps may depend smoothly on state. Local Lipschitz regularity of the resulting right-hand side, regular finite coordinates, admissible initial data and a prescribed regular input give the usual local ODE obligation. These conditions do not prove global continuation, material stability, or that a trajectory remains in the positive-temperature domain. Lower-bounded energy, constitutive stability and any further operating-region invariance are checked for the selected application. Explicit time dependence adds $\partial_t \mathcal{H}$ to equation (73).

7.2 A concrete Maxwell–Lorentz, mechanical and thermal specialization

Select a finite electromagnetic representation on a fixed reference topology with power-dual coordinates. Let d be electric displacement-flux coordinates in coulombs, b magnetic flux coordinates in webers, and C a fixed signed curl/interconnection matrix. Their conjugates e, h have voltage and current units. Matrix entries and coordinate normalization are part of the selected field reduction; these are not point samples or pixel intensities. The skew pair $(C, -C^T)$ is the finite counterpart of curl integration by parts, with boundary power supplied through explicit ports. A compatible field reduction must derive this pair and its boundary terms. Such preservation of Maxwell power structure is developed in UC3: [Brugnoli, Rashad and Stramigioli, Dual field structure-preserving discretization of port-Hamiltonian systems](#).

Add mechanical coordinates q , momenta p , polarization-charge coordinates P_j of the same dimension as d , and their conjugate flux coordinates π_j . Here q is a local mechanical coordinate, distinct from the digital JK word q_n . Set $\delta = d - \sum_j P_j$ and choose

$$\begin{aligned} \mathcal{H} = & \frac{1}{2} p^T M^{-1} p + U(q, s) + \frac{1}{2} \delta^T C_e(q, s)^{-1} \delta + \frac{1}{2} b^T L_b(q, s)^{-1} b \\ & + \sum_{j=1}^m \left(\frac{f_j}{2} \pi_j^T \pi_j + \frac{\omega_j^2}{2 f_j} P_j^T P_j \right). \end{aligned} \quad (75)$$

Here $M = M^T > 0$ is fixed, $C_e = C_e^T > 0$ and $L_b = L_b^T > 0$ are selected electric and magnetic energy matrices, m is finite, and $f_j > 0$, $\omega_j > 0$ are fixed oscillator coefficients. Under the stated flux/charge normalization C_e, L_b, f_j have units F, H, F/s²; π_j has units V s. These finite coefficients are projected/material parameters and are not numerically identified with the continuum coefficients of equation (60) without the actual normalization map. Mixed mechanical dimensions retain the inherited typed-block convention.

The internal energy $U(q, s)$ includes elastic/thermal coupling once. In particular it may be the entropy-variable form of R13's thermoelastic free energy, rather than an independently added second elastic store. Define

$$\begin{aligned} v &= M^{-1} p, & e &= C_e^{-1} \delta, & h &= L_b^{-1} b, \\ V_j &= f_j \pi_j, & w_j &= -e + \frac{\omega_j^2}{f_j} P_j, & T_i &= \partial_{s_i} \mathcal{H}. \end{aligned} \quad (76)$$

For symmetric positive-semidefinite mechanical damping C_m and electric conductance G_e , and $\gamma_j \geq 0$, the nonthermal equations are

$$\begin{aligned} \dot{q} &= v, & \dot{p} &= -\partial_q \mathcal{H} - C_m v + F_{\text{ext}}, \\ \dot{\delta} &= C h - G_e e + i_{\text{ext}}, & \dot{b} &= -C^T e, \\ \dot{P}_j &= V_j, & \dot{\pi}_j &= -w_j - \frac{\gamma_j}{f_j} V_j. \end{aligned} \quad (77)$$

The displayed supply is $v^T F_{\text{ext}} + e^T i_{\text{ext}}$. Additional selected electromagnetic boundary ports are paired by the same power convention. The dissipation channels deposited through equation (72) are exactly

$$D_m = v^T C_m v, \quad D_e = e^T G_e e, \quad D_j = \frac{\gamma_j}{f_j} V_j^T V_j. \quad (78)$$

The curl powers cancel as $e^T \mathbf{C} h - h^T \mathbf{C}^T e = 0$; each mechanical or polarization canonical pair cancels in the same way. The remaining nonthermal power is the supply minus the displayed losses, so the preceding energy and entropy theorem applies without an additional unidentified transfer coefficient.

For the fixed f_j , differentiating $V_j = f_j \pi_j$ gives

$$\dot{V}_j = f_j e - \omega_j^2 P_j - \gamma_j V_j. \quad (79)$$

This is the finite-coordinate Lorentz oscillator. The same P_j enters δ in field energy, supplies the electromagnetic response and receives the work that its damping converts to heat. No separate full “absorption heat” is added on top of D_j . The constitutive origin and continuum energy proof remain those of equations (60)–(61).

7.3 Reciprocal optical force and thermal response

For fixed d, b, P_j, π_j, p, s , matrix inverse differentiation gives the mechanical load contributed by the two field stores:

$$(F_{\text{field}})_a = \frac{1}{2} e^T (\partial_{q_a} C_e) e + \frac{1}{2} h^T (\partial_{q_a} L_b) h. \quad (80)$$

Thus $-\partial_{q_a} \mathcal{H} = -\partial_{q_a} U + (F_{\text{field}})_a$ for the selected fixed M, f_j, ω_j . This derives the reciprocal force from the same geometry/material dependence that changes the field response. It does not insert a generic radiation-pressure formula into every medium. The scalar electroquasistatic restriction is the familiar $F = (e^2/2) \partial_q C_e$, with energy differentiation performed at fixed charge, as derived in [UC4: MIT Electromagnetic Fields, Forces, and Motion, lecture 12, pages 1–2](#). Holding voltage by a source changes the charge and supplies electrical work; one cannot differentiate a fixed-voltage expression as though that source were absent.

Likewise the common thermal derivative is

$$T_i = \partial_{s_i} U - \frac{1}{2} e^T (\partial_{s_i} C_e) e - \frac{1}{2} h^T (\partial_{s_i} L_b) h. \quad (81)$$

The positivity domain applies to this complete expression. Simply prescribing both arbitrary $C_e(q, s)$ and an unrelated $T_i(s_i)$ would overdetermine the thermodynamic constitutive model. As a simple uncoupled thermal store, $U_i(s_i) = c_i T_{i0} \exp[(s_i - s_{i0})/c_i]$ with $c_i, T_{i0} > 0$ gives $\partial_{s_i} U_i = T_{i0} \exp[(s_i - s_{i0})/c_i] > 0$; the exponent is dimensionless. Once field or elastic storage depends on entropy, its derivative must also be included. This identity is a constitutive illustration, not a numerical example or a universal material law.

Material data are often supplied versus temperature. For one thermal coordinate, a twice differentiable Helmholtz energy $\mathcal{F}(z, T)$ gives

$$s = -\partial_T \mathcal{F}(z, T), \quad c_z = -T \partial_T^2 \mathcal{F} > 0, \quad \mathcal{H}(z, s) = \mathcal{F}(z, T(z, s)) + T(z, s)s. \quad (82)$$

At fixed z , $\partial_T s = c_z/T > 0$ permits a local inverse on the declared branch. Differentiation cancels the ∂T terms and yields $\partial_s \mathcal{H} = T$ and $\partial_z \mathcal{H} = \partial_z \mathcal{F}|_T$. For several temperatures the corresponding $-\partial_T^2 \mathcal{F}$ must be nonsingular on the selected branch; positive definiteness supplies local thermodynamic stability of that temperature block. A constitutive inversion still needs its supplied exact finite definition and domain. This connects R13’s free-energy closure to the common entropy state without omitting its reciprocal thermoelastic term.

The force theorem is a theorem about the chosen reduced energy. To represent a moving physical device, q must describe a declared deformation or component coordinate with a valid energy reduction and boundary work. Fixed reference topology with parameterized energy matrices can describe selected small-motion or quasistatic couplings; it is not an arbitrary moving-medium Maxwell model. Unmodeled basis motion, motional electromotive terms, relativistic effects or unrepresented mechanical momentum are not supplied by writing a derivative.

7.4 Field constraints, material modes and exact seed binding

The Maxwell divergence conditions still constrain any proposed field representation. For an unreduced compatible flux description, let $N_e \mathbf{C} = 0$ and $N_b \mathbf{C}^T = 0$ express the boundary-of-boundary identities. If $N_e d = \rho$ initially and the free-charge coordinates obey

$$\dot{\rho} = N_e(-G_e e + i_{\text{ext}}), \quad N_b b = 0 \text{ initially}, \quad (83)$$

then differentiation preserves both constraints. Compatible boundary ports contribute their corresponding charge term. The signs use i_{ext} as an injected electric-flow input; an impressed current leaving the field has the opposite sign. Polarization is already part of d and is not an additional free-charge source.

For the exact finite ODE binding, the selected representation builds these constraints into a regular basis or supplies their explicit elimination and compatible independent inputs, as required by R13. The invariance calculation does not solve incompatible initial data, provide an unspecified boundary condition, or add a differential-algebraic solver to XOP. Modal truncation, constitutive fit and the mapping from finite fluxes to physical fields remain explicit approximation/evidence obligations.

ASA/NA and JK can select a constitutive mode χ , an actuator input or a boundary configuration. In a fixed mode the equations use $\mathcal{H}(z, s; \chi)$. At a mode change with unchanged physical coordinates,

$$W_{\text{switch}} = \mathcal{H}(z, s; \chi^+) - \mathcal{H}(z, s; \chi^-). \quad (84)$$

The value is a required signed supply or export, not proof that an actuator can deliver it. A reset of coordinates or entropy has its own complete before/after balance. A dynamic reporter or hysteretic state affecting energy must likewise have its conjugate exchange and evolution; a declared empirical readout may remain an observation without being counted as an energy store.

The seed specialization records $\mathcal{H}$, its explicit conjugate bodies, the selected matrices, thermal allocations, inputs, initial state and domains in one typed operator graph. Existing scalar arithmetic, finite sums, EXP, and nonsingular SOLVE_LINEAR express the displayed finite bodies when their operands meet the inherited contracts. Gradients in this chapter are mathematical definitions; the specialization supplies their derived finite bodies. Existing DIFF can denote a derivative on its declared domain; it does not by itself provide an automatic differentiation algorithm. The finite ODE notation retains the inherited existence and denotation contract. One-bit words preserve this authoritative graph and its operands.

R13's midpoint balance applies to its fixed linear mechanical specialization. It does not automatically extend to the nonlinear coupled energy (75). Any selected finite update for the combined model needs its own discrete work, heat and entropy identities or stated defects. This chapter adds the integrated physical equations and their continuous proofs; it does not silently claim a new time-step implementation.

8 A finite exact field kernel with work and heat balance

The common continuous energy law needs a separately derived finite update. This section supplies one for a fixed quadratic electromagnetic/mechanical reduction with one thermal store. It extends the linear mechanical midpoint profile to a coupled passive finite field. Temperature-dependent coefficients, nonlinear energy and moving bases retain the continuous model and need their own selected update; they are not covered by this fixed-coefficient proof.

8.1 FIELD-MIDPOINT-QUADRATIC-R1

Let x contain the retained field and mechanical coordinates after resolving algebraic constraints. Declare constant matrices

$$Q = Q^T > 0, \quad J^T = -J, \quad R = R^T \geq 0, \quad A = (J - R)Q, \quad (85)$$

and a constant input map B . The model is

$$\dot{x} = Ax + Bu, \quad E_f(x) = \frac{1}{2}x^T Qx, \quad y = B^T Qx. \quad (86)$$

Each coordinate keeps its own dimensions. In particular $[Q_{ij}] = J/([x_i][x_j])$, the input/output pairing $u^T y$ has power units, and A_{ij} has units $[x_i]/([x_j] \text{ s})$. Positive definiteness refers to the energy form on independent coordinates; it cannot be asserted before removing gauge or other null directions. Fixed constrained Maxwell–Lorentz modes and a fixed positive mechanical stiffness are examples when their energy has this form.

For $h \geq 0$, held discrete input $\bar{u}$ and $\bar{x} = (x^+ + x)/2$, define the undivided midpoint equation

$$x^+ - x = h(A\bar{x} + B\bar{u}). \quad (87)$$

Its complete finite body is

$$\begin{aligned} D_h &= I - \frac{h}{2}A, \\ b_h &= (I + \frac{h}{2}A)x + hB\bar{u}, \\ x^+ &= \text{SOLVE_LINEAR}(D_h, b_h), \quad \bar{x} = \frac{1}{2}(x^+ + x). \end{aligned} \quad (88)$$

For $h = 0$ this gives $x^+ = x$ without a division by time.

Proposition 8.1 (Unique finite solve). *Under the stated hypotheses, D_h is invertible for every $h \geq 0$. If the matrix entries, state, step and input are rational, the output is rational.*

Proof. If $D_h w = 0$, then $w = (h/2)(J - R)Qw$. Multiplication by $w^T Q$ gives

$$w^T Qw = -\frac{h}{2}(Qw)^T R(Qw) \leq 0.$$

Positive definiteness forces $w = 0$, so the square matrix is nonsingular. A finite nonsingular rational linear system has a rational solution. $\square$

8.2 The same update supplies its exact energy ledger

Define the interval work and deposited dissipative energy

$$W = h\bar{u}^T B^T Q\bar{x}, \quad D = h(Q\bar{x})^T R(Q\bar{x}) \geq 0. \quad (89)$$

Symmetry and the midpoint equation give

$$\begin{aligned} E_f(x^+) - E_f(x) &= \bar{x}^T Q(x^+ - x) \\ &= h(Q\bar{x})^T (J - R)Q\bar{x} + h\bar{x}^T QB\bar{u} = W - D. \end{aligned} \quad (90)$$

Thus a lossless unforced instance preserves the selected quadratic field energy, and a passive unforced instance cannot increase it. The skew term exchanges energy between retained coordinates without creating it.

Here R represents only losses thermalized in the selected store. Outgoing radiation or power delivered to an external load belongs in boundary ports or its own store; it is not automatically

deposited as local heat. Connect all of this profile's dissipation to one thermal store of fixed heat capacity $C_\theta > 0$, with $T > 0$, no external heat flow and

$$E_\theta = C_\theta T, \quad S_\theta = C_\theta \log(T/T_\star), \quad T_\star > 0, \quad T^+ = T + D/C_\theta. \quad (91)$$

The arbitrary energy reference is omitted. The temperature remains positive, and direct substitution proves

$$\begin{aligned} \Delta(E_f + E_\theta) &= W, \\ \Delta S_\theta &= C_\theta \log\left(1 + \frac{D}{C_\theta T}\right) \geq 0. \end{aligned} \quad (92)$$

This is a complete finite field/heat update. It deposits the energy D once; adding the same loss again as an independent heater would violate the identity. Heat leaving the selected thermal store requires an additional explicit exchange model. This one-store closure deliberately has no feedback from its temperature into Q, R during the interval.

8.3 Constraints and representation

For a constant linear constraint map L such that $LA = 0$ and $LB\bar{u} = 0$, multiplication of (87) yields $Lx^+ = Lx$. This is a conditional preservation result; an arbitrary modal discretization need not satisfy those identities. For charge-driven Maxwell fields, changing charge and currents need a compatible continuity state and constraint law, rather than a falsely constant constraint.

The template exports (x^+, T^+, W, D) and reads one old snapshot. Its body uses finite scalar operations, typed matrix contraction and the existing `SOLVE_LINEAR` operator. Rational C_θ, T also give a rational next temperature; the logarithm may remain exact syntax if entropy is exported. The packing transform then preserves these operators and operands literally. No PDE opcode, arbitrary-sign oracle or new executable backend is required by the mathematical definition.

The exact residual

$$r_{\text{total}} = E_f(x^+) + C_\theta T^+ - E_f(x) - C_\theta T - W$$

vanishes by the displayed proof. These are discrete work and heat quantities, not a claim that they equal the integrals along the continuous trajectory. Conservation does not establish trajectory accuracy at an arbitrary step size. Changing a coefficient first requires its parameter-work ledger; nonlinear coupling cannot be added to this step while retaining the fixed quadratic proof.

9 Gravity, clocks and station observations from one state

The satellite specialization shares the orbit, gravitational coefficients, station motion, frame realization and clock parameters with its observation map. This closes a specific gap in R11: its orbital station query is simultaneous geometry, whereas its live GPS query uses emission and reception events. The following equations define their common physical interface.

9.1 A source potential with explicit finite exterior models

For an isolated Newtonian mass density $\rho_m \geq 0$, fix the potential zero at infinity. In SI units the source law and its boundary realization are

$$\nabla^2 \Phi = 4\pi G \rho_m, \quad \Phi(x) = -G \int \frac{\rho_m(x')}{\|x - x'\|} d^3 x', \quad g = -\nabla \Phi. \quad (93)$$

Outside the source $\nabla^2 \Phi = 0$. Boundary conditions or measured exterior coefficients are required; the Poisson equation alone does not identify them. [\[GT1: MIT, potential theory\]](#)

Let $R(t) \in SO(3)$ map Earth-fixed components to a common geocentric, nonrotating spatial chart. For $p = R^T r$, $s = \|p\| > a_E > 0$ and $\mu_E > 0$, one closed exterior choice is

$$\Phi_E(t, r) = -\frac{\mu_E}{s} \left[1 - J_2 \left(\frac{a_E}{s} \right)^2 P_2(p_z/s) \right], \quad P_2(u) = \frac{3u^2 - 1}{2}. \quad (94)$$

$J_2 = 0$ selects the monopole. Alternatively select the finite R11 solid-harmonic body with $\Phi_E = -U_{R2}$, retaining its normalization, degree, coefficient epoch and exterior domain. These are alternative Earth fields; adding them would count the monopole twice. $\mu_E = GM_E$ is a gravitational parameter with a declared realization, not an exact SI defining constant. Arbitrary fitted coefficients need not correspond to a nonnegative physical density; consistency with the selected source and coefficient uncertainty is a separate obligation.

For a point body at geocentric $r_b(t) \neq 0$, the appropriate geocentric *tidal* potential and its force are

$$\Phi_b^{\text{tid}}(t, r) = -\frac{\mu_b}{\|r_b - r\|} + \frac{\mu_b}{\|r_b\|} + \frac{\mu_b r_b \cdot r}{\|r_b\|^3}, \quad (95)$$

$$-\nabla_r \Phi_b^{\text{tid}} = \mu_b \left(\frac{r_b - r}{\|r_b - r\|^3} - \frac{r_b}{\|r_b\|^3} \right), \quad r \neq r_b. \quad (96)$$

The constant and linear terms remove the external potential and acceleration at the geocenter. Set $\Phi_{\text{geo}} = \Phi_E + \sum_b \Phi_b^{\text{tid}}$ and

$$\dot{r} = v, \quad \dot{v} = -\nabla_r \Phi_{\text{geo}} + a_{\text{ng}} + a_{1\text{PN}}. \quad (97)$$

a_{ng} binds the selected radiation-pressure, thrust or other physical force body and its parameters. Set absent terms to zero explicitly. The existing R11 spherical-Earth a_{GR} may bind $a_{1\text{PN}}$; it is a specified post-Newtonian correction, not the whole relativistic force model. All retained force choices remain subject to their source/model error.

9.2 Frame acceleration and gravitational energy

For a general Newtonian inertial chart $x = o(t) + R(t)p$, define $\Omega = R^T \dot{R}$ and $\Omega z = \omega \times z$. Differentiating twice gives

$$\dot{x} = \dot{o} + R(\dot{p} + \omega \times p), \quad (98)$$

$$\ddot{p} = R^T(\ddot{x} - \ddot{o}) - 2\omega \times \dot{p} - \dot{\omega} \times p - \omega \times (\omega \times p). \quad (99)$$

Thus a rotating-frame propagation includes translation, Coriolis, Euler and centrifugal terms. In (97) the geocenter acceleration has already been removed by the tidal potential; it is not subtracted a second time. The identity follows from $R^T R = I$, which also makes Ω skew. Independently fitted matrix entries in R11 do not establish this identity; an R14 frame realization must prove it or retain a bounded frame discrepancy. [\[GT2: MIT, rotating axes\]](#)

For the Newtonian part and constant satellite mass m , differentiation yields

$$\frac{d}{dt} \left(\frac{m\|v\|^2}{2} + m\Phi_{\text{geo}}(t, r) \right) = m v \cdot a_{\text{ng}} + m \partial_t \Phi_{\text{geo}}. \quad (100)$$

This equation assumes $a_{1\text{PN}} = 0$; otherwise its work must be retained and a relativistic energy model selected for a relativistic conservation claim. Moving external sources exchange energy through $\partial_t \Phi_{\text{geo}}$. It is not heat or arithmetic drift. An Earth-fixed constant field can still be time dependent in the nonrotating chart. A finite integration operator needs its own energy identity; the continuous balance does not certify RK4.

9.3 Proper time and the same gravitational state

Take t as TCG and the spatial/force data as compatible geocentric coordinates. In the weak-field, slow-motion regime the leading proper-time rate of an ideal clock on trajectory $r_A(t)$ is

$$\frac{d\tau_A}{dt} = 1 + \frac{\Phi_{\text{geo}}(t, r_A) - \|v_A\|^2/2}{c^2} + \epsilon_{\text{clk},A}(t). \quad (101)$$

$\epsilon_{\text{clk},A}$ accounts for omitted higher-order/model terms; it is not an arbitrary control input. The negative-potential convention fixes the sign: a stationary clock deeper in a monopole field has a smaller rate. The tidal potential above, rather than the full absolute Sun/Moon potential, is appropriate to this geocentric convention. A barycentric clock model uses its own coordinate transformation and potential. [GT3: IERS, chapter 10, equations 10.6–10.9]

For a scaled coordinate $T = \alpha t + \beta$, $\alpha > 0$, the chain rule gives $d\tau_A/dT = (d\tau_A/dt)/\alpha$. TT uses $\alpha = 1 - L_G$, with exact $L_G = 6.969290134 \times 10^{-10}$; station, orbit and gravitational parameter values must be compatible with the chosen coordinate scale. GPST tags require their declared realization and mapping. A stored rational timestamp does not prove a physical clock exact or turn (101) into R11's rational time-conversion template.

The readout is modeled by $C_A(t) = t + \delta_A(t)$, where a declared oscillator and steering model connects C_A to τ_A . Clock initial offset, intrinsic frequency error and steering belong to the shared uncertainty state. A broadcast clock product may already contain physical corrections: apply its documented convention once, rather than adding (101) again. [GT4: ESA, relativistic clock correction]

9.4 Causal reception, range and pointing

For satellite emission at t_e and station reception at $t_r > t_e$, define

$$d = r_s(t_e) - r_g(t_r), \quad \rho = \|d\| > 0, \quad t_r - t_e = \frac{\rho}{c} + \Delta_{\text{grav}} + \Delta_{\text{med}}. \quad (102)$$

This is a two-event equation in one coordinate chart. For the leading static Earth monopole propagation correction, let $s_e = \|r_s(t_e)\|$, $s_r = \|r_g(t_r)\|$ and use

$$\Delta_{\text{grav}} = \frac{2\mu_E}{c^3} \log \frac{s_e + s_r + \rho}{s_e + s_r - \rho}. \quad (103)$$

It requires $s_e + s_r > \rho$ and a nonocculted exterior ray in the weak-field domain. Δ_{med} binds a frequency-dependent propagation model appropriate to the measured signal, or is zero in the declared vacuum model. Higher gravity terms, bending, atmospheric error and multipath remain in the observation discrepancy where omitted. [GT5: IERS, chapter 11, ranging equation 11.17]

For the flat vacuum choice $\Delta_{\text{grav}} = \Delta_{\text{med}} = 0$ and fixed t_r , put $F(u) = u + \|r_s(u) - r_g(t_r)\|/c - t_r$. On a collision-free interval with $\|\dot{r}_s\| < c$, $F'(u) = 1 + \hat{d} \cdot \dot{r}_s/c > 0$. A bracket $F(a) \leq 0 \leq F(b)$ therefore specifies a unique emission time. Additional delays require a corresponding monotonicity bound; a fixed number of correction iterations is not identical by definition to this root.

When positions are supplied in Earth-fixed components and both charts share the geocenter, the retarded geometric vector in reception axes is

$$d_E = R(t_r)^T R(t_e) r_{s,E}(t_e) - r_{g,E}(t_r), \quad z = L_g(t_r) d_E = (E, N, U)^T. \quad (104)$$

$L_g \in SO(3)$ is the declared local ENU rotation. Then $\text{az} = \text{atan2}(E, N)$ and $\text{el} = \text{atan2}(U, \sqrt{E^2 + N^2})$ on their nonsingular domains. This product includes Earth rotation during flight; adding a second Sagnac correction would duplicate it. These are retarded geometric angles. Physical

detector pointing additionally binds aberration, ray bending, refraction and mounting terms when required. [GT6: ESA, reception-frame coordinates]

For code readouts C_r, C_s , a corresponding pseudorange observation is

$$P = \rho + c(\Delta_{\text{grav}} + \Delta_{\text{med}}) + c[\delta_r(t_r) - \delta_s(t_e)] + d_p + \eta_p. \quad (105)$$

Here d_p is the remaining modeled discrepancy/bias and η_p the declared measurement error. Orbit coefficients, both clock states, station motion, Earth orientation and propagation parameters enter the joint uncertainty set of Section 11. Their correlations are preserved. Changing gravity, time or frame in one branch changes every dependent branch. Literal one-bit encoding preserves these equations and their operands; a physical tolerance still requires the bound in Equation (128) for the chosen output and interval.

10 An atomic-time observation, literal latch and domain bindings

An actual NIST internet-time observation is included as a fixed input to this formalization. Its literal words provide a reproducible event anchor for the physical models. They do not cause a running clock, calibrate an oscillator, or supply the initial state of every physical system at that event.

The SI atomic reference is the unperturbed ground-state hyperfine transition frequency of cesium-133:

$$\Delta\nu_{\text{Cs}} = 9192631770 \text{ Hz}, \quad 1 \text{ s} = 9192631770 / (\Delta\nu_{\text{Cs}}). \quad (106)$$

The defining value is exact; a realization has measurement uncertainty. [CL4: BIPM, SI defining constants] UTC(NIST) uses an ensemble of cesium-beam and hydrogen-maser clocks calibrated against NIST's primary frequency standard. Its internet service distributes that atomic reference; this capture does not install an atomic oscillator in aTOMos. [CL5: NIST, UTC(NIST) realization]

10.1 The observed words and their exact rational coordinate

The recorded exchange with `time.nist.gov`, endpoint `132.163.96.4:123`, used a version-4 request and returned a version-3 response, mode 4, stratum 1, leap indicator 0 and reference ID NIST. The complete request, reply and local acquisition stamps are preserved in `source/clock_capture/nist_capture.json`. The server transmit field is

$$w_{\text{tx}} = 17173213285462662115 \quad \text{as an unsigned 64-bit word}, \quad e = 0. \quad (107)$$

The era e is bound by the recorded contemporary-date context; the packet does not carry it. The reply's SHA-256 is

$$26a429327733f35d8b67abfb9bbb33f54fae9fa590c4ba0e5a7947fe6946c3e4.$$

For the 32-bit seconds and fraction fields $s, f \in \{0, \dots, 2^{32} - 1\}$, the mathematical decoder is

$$w = 2^{32}s + f, \quad q_N = e \cdot 2^{32} + s + \frac{f}{2^{32}}, \quad q_U = q_N - 2208988800. \quad (108)$$

Here q_N uses the 1900 NTP epoch and q_U the corresponding 1970 civil coordinate. Every operation is integer/rational; no floating conversion is needed. The observed word gives the reduced rational

$$q_U = \frac{7685678632232377315}{4294967296}, \quad (109)$$

whose UTC display, truncated to microseconds, is `2026-09-15T08:38:22.859456Z`. The display is not the authoritative value. The NTP fixed-point format and era distinction follow [CL1: RFC 5905, section 6].

NTP's coordinate is affected by leap seconds; differences across leap events are not automatically elapsed SI durations. A complete mapping binds the era evidence, leap table, applicable interval and any ambiguity/smear policy. [CL2: RFC 8877, NTP timestamps] NIST documents its positive-leap convention as repetition of the final second's binary timestamp. [CL3: NIST Internet Time Service] This convention is part of the source identity; replacing the provider does not silently inherit it. The earlier Cloudflare capture is retained separately as acquisition provenance; the NIST record is the selected latch.

10.2 Complete nominal mapping at the captured epoch

The retained `source/clock_capture/leap-seconds.list` has last transition $q_N = 3692217600$ with TAI–UTC offset 37 seconds and declared expiry $q_N = 4023129600$. Its SHA-256 is

db5a895f16853b03bfc865e8d68f9fc8710ef1740e3400c701cd46a5bbbc3433.

For the captured non-leap instant in this bound interval, the nominal conversion body is explicitly

$$\begin{aligned} Q_{\text{TAI}} &= q_U + 37, \\ t_{\text{GPST}} &= q_U - 315964800 + (37 - 19), \\ Q_{\text{TT}} &= q_U + 37 + \frac{4023}{125}, \\ Q_0 &= 220924800 + \frac{4023}{125}, \quad t_{\text{TCG}} = \frac{Q_{\text{TT}} - Q_0}{1 - L_G}. \end{aligned} \quad (110)$$

GPST is elapsed seconds from 1980-01-06 00:00:00 GPST. Q_{TAI} and Q_{TT} use 1970-01-01 00:00:00 calendar-coordinate origins on their respective scales. t_{TCG} is elapsed TCG seconds from 1977-01-01 00:00:32.184, when TT and TCG are defined equal; Q_0 is that event's TT coordinate and $L_G = 6.969290134 \times 10^{-10}$ is exact. These are finite rational expressions for the nominal scales. The leap file is a dated dependency, not authority for an unannounced future leap or an ambiguous repeated second. UTC(NIST)–UTC and GNSS realization differences remain physical uncertainty. [CL6: IANA, retained leap table] [GT3: IERS, time coordinates]

10.3 What the clock observation establishes

The timestamp denotes the server's transmit event. Its observed local monotonic round-trip interval is the exact recorded count $127594000/10^9$ seconds under the local timer convention. Neither that count nor stratum, precision or dispersion metadata certifies a UTC error bound. The UDP reply was not cryptographically authenticated. The host clock was not changed and no oscillator calibration was performed.

Let q_* be the nominal converted source-event coordinate and write

$$t_{\text{tx}} = q_* + \epsilon_s, \quad t_{\text{rx}} = t_{\text{tx}} + d_{\leftarrow}, \quad d_{\leftarrow} \geq 0. \quad (111)$$

Source clock error ϵ_s , inbound delay $d_{\leftarrow}$, local readout error and conversion/realization error enter one joint uncertainty set. An interval for any of them needs supporting assumptions or calibration. One exchange does not determine asymmetric one-way delays or prove source authenticity. Literal retention preserves the observation actually received.

For a complete four-stamp exchange on compatible, unfolded coordinates, the conventional offset and round-trip expressions are

$$\hat{\theta} = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}, \quad \hat{d} = (T_4 - T_1) - (T_3 - T_2). \quad (112)$$

These expressions can remain exact rationals. Interpreting $\hat{\theta}$ as the physical offset requires a clock/path model; unequal inbound and outbound delays contribute an offset error. [CL1: RFC 5905, section 8]

10.4 Immutable XOP snapshot and a one-time latch

Define a finite typed tuple C containing both raw packet words, source identity, era and its evidence, acquisition stamps, protocol fields, the rational coordinate and the complete time-mapping dependency identities. The packet words use `Bits(384)` for this 48-octet capture. Other fields use existing integer, rational, Boolean and fixed structural types; the binding fixes their widths and ordering. No new timestamp opcode is needed.

The completed capture may be a closed dependency value. A newly acquired capture instead enters through a sample port. Every consumer at a logical step reads the same immutable old-state/input snapshot. The physical mapping is an imported complete pure template, with its declared domain, rather than an instruction to consult the network again. The materialized `source/clock_capture/anchor_state.json` records the selected NIST observation and nominal rational mappings. The equations here specify its XOP binding; no running XOP clock executor is claimed.

Let a_n be an accepted-record pulse, v_n the admission predicate for its typed record, ℓ_n the old latched Boolean, and H_n the stored tuple. The explicit one-time latch is

$$\begin{aligned} e_n &= \text{IF}(\ell_n, \text{false}, \text{IF}(a_n, v_n, \text{false})), \\ \ell_{n+1} &= \ell_n \vee e_n, \quad H_{n+1} = \text{IF}(e_n, C_n, H_n). \end{aligned} \quad (113)$$

Initially $\ell_0 = \text{false}$ and H_0 is a fixed closed placeholder of the declared type, which has no observed-clock meaning while unlatched. For defined inputs the pulse equals $a_n \wedge v_n \wedge \neg \ell_n$. The displayed lazy form demands no new record, admission check or pulse once latched, and demands no record when an unlatched acceptance pulse is false. Both next roots commit together. Once ℓ_n is true, e_n is false at every later step and induction proves $H_{n+k} = H_n$ for all $k \geq 0$. Thus future internet replies cannot alter this seed's latched anchor. If an electrical rising edge supplies acceptance, a declared old Boolean b_n gives $a_n = c_n \wedge \neg b_n$ and $b_{n+1} = \text{IF}(\ell_n, b_n, c_n)$ in the same commit. Thus the optional edge memory also demands no new level once latched. A later anchor is a separately versioned capture, not an implicit rewrite.

The raw sample time and canonical-time value remain distinct: R11 tag 5 exposes the source rational and tag 6 applies the complete `TimeConvention` mapping to GPST-SI. An unresolved era or leap branch produces an explicit non-value, not a guessed instant. Closing the captured record as a seed dependency also closes its mapping table and source interval. No live lookup is hidden in a pure expression.

10.5 One anchor, explicit time maps for each physical domain

Let t_0 denote the nominal latched event on the selected canonical scale. Each domain uses a declared map $t_d = \chi_d(t)$ and its own state, frame, boundary conditions and observation history. For a differentiable increasing map, the equation transforms by the chain rule:

$$\frac{dx_d}{dt} = \frac{d\chi_d}{dt} F_d(\chi_d(t), x_d, u_d), \quad x_d(t_0) = x_{d,0}. \quad (114)$$

The supplied state $x_{d,0}$ must refer to that event. An old observation cannot become a current state merely by changing its timestamp.

Domain	Explicit binding to the latched event
Electromagnetism and photonics	$E(x, t), B(x, t), P_j(x, t)$ and their source/boundary histories use the mapped field time. A carrier phase $\varphi(t) = \varphi_0 + \int_{t_0}^t \omega(s) ds$ also needs its phase/frequency reference; NTP does not measure optical phase.
Mechanics and control	$q(t_0), p(t_0)$, attitude, loads and actuator events retain their initial-event identity. ASA/NA and JK transitions act through their explicit physical ports.
Thermal and material history	$T(t_0)$, internal variables and accumulated work/heat require the actual state/history binding. A clock latch does not reset energy or entropy.
Gravity and orbit	The GPST/TT/TCG mapping, compatible frame/force parameters and station/satellite events remain those of Section 9. A coordinate-time conversion does not measure proper clock time.
Sensors and calibration	Acquisition endpoints, integration windows, latency and calibration age use mapped recorded events, with the shared clock error in the observation uncertainty.
Digital state and replay	The integer transition index n is causal order. A time $t_n = t_0 + \sum_{j < n} h_j$ requires explicit durations h_j ; n alone is not seconds.

A common event does not require equal sample rates or spatial rasterization. Field expressions may retain their continuous variables and exact implicit denotation. A selected finite mode basis or finite step remains a separately specified approximation with its own discrepancy. Clock anchoring supplies the temporal relationship; it does not turn a finite model into every physical domain or remove source, state or model uncertainty.

11 Measurement, constants and output tolerances

The existing XOP dimension, frame, sample and dependency types already carry the information needed here. This chapter adds a physical interpretation and conditional output certificates, without adding a measurement oracle or a new packet format. An exactly stored indication, an exact nominal model and the physical quantity being inferred remain distinct. The purpose is operational: preserve what is known, keep shared uncertainties coupled, and determine what an output can claim within a selected application tolerance.

11.1 Exact defining constants and measured coefficients

Five SI defining constants connect the present time, optical, electrical and thermal profiles. Their numerical values are fixed by the SI; unit realizations and measurements made with them still have uncertainty. The following dimension vectors use the inherited order $(L, M, T, I, \Theta, N, J_{\text{lum}})$. [\[ME1: BIPM defining constants\]](#)

Constant	Exact SI quantity	XOP dimension vector
$\Delta\nu_{\text{Cs}}$	9 192 631 770 Hz	(0, 0, -1, 0, 0, 0, 0)
c	299 792 458 m s ⁻¹	(1, 0, -1, 0, 0, 0, 0)
h	6.626 070 15 × 10 ⁻³⁴ J s	(2, 1, -1, 0, 0, 0, 0)
e	1.602 176 634 × 10 ⁻¹⁹ C	(0, 0, 1, 1, 0, 0, 0)
k_{B}	1.380 649 × 10 ⁻²³ J K ⁻¹	(2, 1, -2, 0, -1, 0, 0)

The decimal coefficients are finite rational literals, reduced canonically by the XOP grammar. For example, the numerical coefficient of h is 662607015/10⁴² before reduction. A binary floating-point approximation is unnecessary. The elementary charge e is positive; the charge of an electron is $-e$. The namespace distinguishes it from the Boolean accepted-event bit.

The gravitational constant G and fine-structure constant α_{fs} are measured quantities, with dimensions $L^3M^{-1}T^{-2}$ and 1, respectively. A specialization that needs them binds a dated value, uncertainty and provenance; this core does not hard-code a new estimate. [ME2: NIST CODATA table] In present SI, vacuum permeability and permittivity are also not independent exact decimal constants. For positive α_{fs} , retain their common dependency:

$$\mu_0 = \frac{2\alpha_{\text{fs}}h}{e^2c}, \quad \epsilon_0 = \frac{e^2}{2\alpha_{\text{fs}}hc}, \quad \mu_0\epsilon_0 = \frac{1}{c^2}. \quad (115)$$

Here $[\mu_0] = MLT^{-2}I^{-2}$ and $[\epsilon_0] = M^{-1}L^{-3}T^4I^2$. Multiplication cancels the shared α_{fs} exactly. Replacing the two occurrences by unrelated uncertain inputs would lose that constraint. The former exact assignment $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ is not the present SI definition. [ME3: SI Brochure, 9th edition, 2026, section 2.3.2]

11.2 An observation is a modeled acquisition

Let $x(t)$ be the selected physical state, θ its sensor/calibration parameters, and m the transduction law into an analogue indication. An averaging acquisition over $[t_-, t_+]$ has the model

$$M[x, \theta, w] = \int_{t_-}^{t_+} w(t) m(x(t), \theta, t) dt, \quad w(t) \geq 0, \quad \int_{t_-}^{t_+} w(t) dt = 1, \quad (116)$$

$$y = Q_{\Delta}(M[x, \theta, w] + d_M + \eta). \quad (117)$$

The weighting w has dimension T^{-1} ; y, M , the model discrepancy d_M and analogue noise η have the same indication dimension. For a counter or accumulated dose, use its declared accumulation law instead of a normalized average. An instantaneous sample is a separately defined profile, not an implicit substitution of the interval endpoint for the integral. Each integral requires finite bounds and its stated integrability conditions. The profile supplies finite function templates or explicit external bindings for the trajectory and weighting; it cannot encode an arbitrary unknown continuous signal as a fully known finite literal.

Q_{Δ} includes its step, offset, ties rule, range and saturation behavior. For an unsaturated uniform nearest quantizer with step $\Delta > 0$, put $v = M + d_M + \eta$. Then

$$y = M + d_M + \eta + q, \quad q = Q_{\Delta}(v) - v, \quad |q| \leq \Delta/2. \quad (118)$$

This is a deterministic bound, not an assumption that q is uniform, independent or newly random at every sample. Saturated codes require the actual preimage of the saturation law; the half-step bound no longer follows. The stored integer and its rational unit conversion can be exact while the corresponding physical indication occupies a range.

The measurement model also declares its valid specimen, operating conditions and reference conditions. A discrepancy bound is supplied evidence or a stated hypothesis, never a number

generated merely by writing d_M . Unrecognized effects are not covered automatically. This makes explicit the model scope emphasized by JCGM GUM-6, sections 6–7. [ME4: [Developing and using measurement models](#)]

11.3 Calibration and joint uncertainty

Let p collect the finite parameters of the chosen profile: initial state, calibration, clock and frame realization, input history parameters and relevant residuals. A joint admissible set $\mathcal{U}$ retains their actual shared dependencies. With observation map F built from (117), the data-compatible set is

$$\mathcal{U}_y = \{p \in \mathcal{U} : F(p) = y\}, \quad \mathcal{Z}_y = \{g(p) : p \in \mathcal{U}_y\}. \quad (119)$$

For multiple records, impose all their observation equations on the same shared calibration parameters. A repeated common offset does not become a new independent offset. If the set is proved empty, the declared model, admissible bounds and observations are inconsistent; empty-set containment must never be used to certify an output. If solving it is unresolved, retain the unresolved status. Neither conclusion identifies which physical assumption failed.

A simple calibration model $y^{\text{cal}} = A\theta + \varepsilon$ illustrates what equations alone can establish. With known design matrix $A \in \mathbb{R}^{m \times r}$, for consistent noiseless data the parameters are unique exactly when

$$\text{rank } A = r. \quad (120)$$

For a chosen positive-definite covariance Σ and full column rank, the generalized least-squares expression is

$$\hat{\theta} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} y^{\text{cal}}. \quad (121)$$

The inverse exists since $z^T A^T \Sigma^{-1} A z = (Az)^T \Sigma^{-1} (Az) > 0$ for $z \neq 0$. This proves the algebraic expression's domain, not a calibration accuracy. Uncertain reference settings make A uncertain and must enter the joint model. For a differentiable nonlinear calibration, a full-column-rank Jacobian permits a local regular inverse by selecting independent observation components. It does not give global uniqueness. Rank deficiency alone does not decide nonlinear identifiability.

An alternative statistical profile supplies a joint distribution or covariance with its basis. For a differentiable output map and its linearization $J = Dg(\hat{p})$,

$$\Sigma_z \simeq J \Sigma_p J^T, \quad u^2(z_i) \simeq \sum_{j,k} J_{ij}(\Sigma_p)_{jk} J_{ik}. \quad (122)$$

Cross terms retain correlations. This is exact for an affine map with the stated covariance; in general it is a first-order approximation and a standard uncertainty is not a worst-case bound. [ME5: [JCGM 100, sections 5.1–5.2](#)] Significant nonlinearity can also affect the output estimate itself; exact evaluation of $g(\hat{p})$ does not make it the distributional mean. [ME6: [JCGM 100, Amendment 1:2026](#)]

11.4 Acquisition time and realized coordinates

R12's WI record, immutable registry and session consumption rules remain unchanged. The calibration identifier resolves complete calibration content; its presence is not proof of freshness or traceability. Add the chosen measurement model, acquisition weighting, quantizer and joint-uncertainty definition to that existing dependency closure.

For recorded source times $\tau_{\pm}$, a declared affine clock model is

$$t_{\pm} = a\tau_{\pm} + b + \delta t_{\pm}, \quad a > 0. \quad (123)$$

The same a, b appear at both endpoints and in every record covered by that calibration. The physical acquisition interval, conversion validity interval and availability time remain distinct.

Subtracting endpoints cancels the shared offset b ; treating it as two independent offsets would miss this exact dependency. The uncertainty of a and endpoint residuals remains.

The analogous physical frame map is

$$x_B(t) = R_{BA}(t)x_A(t) + o_{BA}(t), \quad R_{BA}^T R_{BA} = I, \quad \det R_{BA} = 1. \quad (124)$$

The XOP frame identity specifies the convention; R_{BA} and o_{BA} specify its realization. A shared uncertain mounting is retained once in p , and an interval enclosure of its entries must retain, or conservatively enclose, the rotation constraints. A nominal matrix is not made physically aligned merely by encoding it exactly. Coordinate values from different acquisition times require the selected dynamical/time-transfer model before they can represent simultaneous geometry.

11.5 A bounded output certificate

The core needs a reusable rule for deciding a functional tolerance without inventing a universal numerical tolerance. Let K be a nonempty compact convex set in the finite parameter coordinates; require $\hat{p} \in K$ and a nonempty $\mathcal{U}_y \subseteq K$. The output g_i is continuously differentiable on an open neighborhood of K . Suppose the following finite bounds have been established for all $p \in K$:

$$|p_j - \hat{p}_j| \leq r_j, \quad \left| \frac{\partial g_i}{\partial p_j}(p) \right| \leq L_{ij}. \quad (125)$$

L_{ij} has output-unit per parameter-unit dimensions. No covariance or independence assumption is used. By integrating the derivative along the segment from $\hat{p}$ to p ,

$$g_i(p) - g_i(\hat{p}) = \int_0^1 \sum_j \frac{\partial g_i}{\partial p_j}(\hat{p} + s(p - \hat{p}))(p_j - \hat{p}_j) ds, \quad (126)$$

$$|g_i(p) - g_i(\hat{p})| \leq \sum_j L_{ij} r_j. \quad (127)$$

The compact domain stays clear of undefined divisions, branch changes or other points where the derivative hypotheses fail. A box can conservatively enclose a correlated set; using the original joint constraints can tighten the result. Piecewise or discrete outputs use direct image/branch enclosure instead of differentiating through a switch.

To apply the certificate to a physical result, assume its actual parameter vector lies in $\mathcal{U}_y$. Nonemptiness alone does not establish that physical inclusion. Suppose the intended physical output is $z_i^{\text{phys}} = g_i(p) + d_i^{\text{out}}$ with $|d_i^{\text{out}}| \leq b_i$, and the delivered numerical output obeys $|z_i^{\text{del}} - g_i(\hat{p})| \leq \rho_i$. Then

$$|z_i^{\text{phys}} - z_i^{\text{del}}| \leq B_i := \sum_j L_{ij} r_j + b_i + \rho_i. \quad (128)$$

This triangle-inequality certificate separates input/calibration uncertainty, output-model discrepancy and representation/evaluation error. Effects already included in p are not counted again in b_i . The XOP exact expression can be retained with no numerical rendering error in its denotation; a finite rendering needs its own ρ_i certificate. Existing binary64 execution is not assigned $\rho_i = 0$ by this new formalization.

For a requested absolute error tolerance $\tau_i \geq 0$, the condition $B_i \leq \tau_i$ is sufficient on the stated domain and acquisition interval. For an allowed physical operating band $[\ell_i, u_i]$, sufficient containment is

$$z_i^{\text{del}} - B_i \geq \ell_i, \quad z_i^{\text{del}} + B_i \leq u_i. \quad (129)$$

Failure to prove this containment does not prove violation: a conservative enclosure may overlap both regions. A wholly disjoint enclosing interval proves exclusion; overlap remains unresolved. This is the general form of R12's bounded bridge and timing decisions. It certifies the declared quantity and interval, not every later state or a different physical application.

11.6 Minimal specialization record

Each specialization binds only: the measurand and conditions; the complete measurement/calibration expressions and actual source records; the unit, frame and acquisition-time realizations; the joint uncertainty or statistical model; and the requested output tolerance with its scoped evidence. The certificate binds those same dependency versions. Changing a calibration, clock, model basis, duration or tolerance requires reevaluating the affected claim, not changing the one-bit packing format.

These are metadata and template/dependency obligations under XOP-R1. Set images, universal bounds and physical evidence are not secretly new executable opcodes. A backend may return UNKNOWN, MISSING_INPUT or RESOURCE_LIMIT when it cannot establish an obligation. The bridge, thermal and optical expressions remain useful before that backend exists.

Metrological traceability is a property of a measurement result supported by its calibration chain and uncertainty; source hashing and exact arithmetic preserve provenance but do not supply that chain. Fitness for a particular use additionally depends on the required uncertainty or tolerance. [\[ME7: NIST traceability policy\]](#) The additions here make that distinction actionable through (128), rather than adding unrelated domain rules.

12 Specialization, proof scope and delivered authority

The unified foundation now contains actual common bodies, rather than only shared names: literal word feedback, exact operator storage, continuous physical balances, a coupled energy function, finite work/heat updates, gravity/time geometry and an acquisition/uncertainty contract. The latched NIST observation supplies one real time reference whose identity can be shared by those domains. It supplies no missing initial field, material coefficient, station coordinate or detector calibration.

12.1 The minimal specialization record

An application supplies the record Π defined in Section 3, its actual equation bodies, and the following bindings. Each dependency is included only when the selected outputs use it.

Binding	Required content
State	Types, frames, initial values or acquired initial-state requirements, and the complete exact roots retained after a step
Time	Source event and epoch, scale map, acquisition and availability intervals, clock uncertainty and the logical step policy
Physics	Applicable laws, constitutive energy/response, geometry, independent boundary data, sources and admitted domain
Coupling	Shared energy, reciprocal derivatives, power ports, heat destinations, and parameter-switch work
Evolution	Continuous solution declaration or specified finite step, existence/domain obligations and its own invariants or defects
Observation	Actual transduction and projection, unit/clock/frame realization, correlated uncertainty, requested output and tolerance
Realization	Selected software or physical encoding, resource bounds and the evidence supporting the claims actually made

An omitted unused domain requires no surrogate equation. An omitted required input remains an open dependency. This makes the formalized core reusable without adding a catalogue of unrelated laws. Conservation, well-posedness, physical correspondence and accuracy are distinct

obligations, each tied to the equations and observations to which it applies.

12.2 What the equations establish

The literal bit-plane transform is invertible on complete canonical seeds. The delayed typed graph has a unique next state whenever its demanded operations are defined. The common continuous field/thermal model has the derived energy and entropy balances on its admitted domain. The mechanical and quadratic-field midpoint profiles have their own exact discrete work identities and unique solves under their fixed passive assumptions. The gravity and acquisition equations tie clocks, coordinates and physical outputs together. The measurement bound gives a sufficient tolerance condition when the actual parameters and model/output errors meet its premises.

These results are mathematical consequences within the specified models. The source observations support their recorded values and provenance. No new executable XOP engine, proof assistant, manufactured photonic arithmetic, general physical performance or improved satellite accuracy is claimed. The logical reviews are internal independent reviews; physical correspondence requires the selected application's observations and characterization.

12.3 One current document in the main repository

The main-repository PDF and the release PDF have the same bytes. The editable master is `docs/unified.tex`. The word/operator catalog is included in the appendix; the corresponding canonical source is `formal/OPERATOR_CATALOG.md`. Domain derivations and exact clock binding are in `formal/`. Raw NIST time data and the IANA leap table are in `source/clock_capture/`. The source-binding record identifies the older documents and the exact source material used in this consolidation.

The review record separates mathematical review, the requested clock observation, visual PDF inspection and artifact byte integrity. The release manifest and archive preserve the delivered sources. Older releases remain available with their original profiles, definitions and historical evidence.

A Normative XOP-R1 grammar and operator registry

This chapter embeds the literal, type, wire and operator contract from `formal/OPERATOR_CATALOG.md`. It supplies the actual fields and numbered operators needed to read the proposed profile without another file. The mathematical denotations and conditional proofs are in Section 1; the application connection is in Section 2. This is a specification, not an implemented parser, exact evaluator or proof checker. Unassigned tags and opcodes are rejected; the new grammar does not reinterpret the preserved R10 format.

A.1 Sorts, outcomes and canonical byte primitives

The scalar embeddings are explicit and value preserving: $\text{Int} \hookrightarrow \text{Rat} \hookrightarrow \text{Alg} \hookrightarrow \text{RealTerm}$. Booleans and finite words are separate types. `RealTerm` carries finite syntax with a possibly partial real denotation; it is not a literal for an unspecified arbitrary real number. Structural types carry their shapes. A `Dim` scalar additionally carries seven rational SI exponents. A `Frame` annotation encodes nominal coordinate identity; prose alone cannot supply it. Angle conventions remain declared interface metadata.

Outcome	Meaning
VALUE	An exact literal with evidence of its construction.
TERM	Retained typed syntax, assumptions and outstanding obligations.
UNDEFINED	A violated domain condition has been established.
UNKNOWN	A required comparison, domain or existence claim is undecided.
MISSING_INPUT	A required dependency or sample is unavailable.
RESOURCE_LIMIT	The declared memory, time or proof-search limit is reached.

These outcomes are separate from scalar values. Unknown or exhausted resources mean neither false nor zero. A retained term is not a proved real value. `COMPARE` is a decision-service interface returning justified LT, EQ, GT, or UNKNOWN; it is not an arithmetic oracle.

Concatenation in the grammar has no implicit alignment. The byte primitives are:

Primitive	Encoding
U(n)	Minimal unsigned base-128 integer: seven payload bits per octet, least significant digit first; high bit one on every nonterminal octet and zero on the last. Zero is one zero octet. A multi-octet encoding cannot end in a zero payload. Counts have no format-level 64-bit ceiling.
O(b)	One octet, $0 \leq b \leq 255$.
B(bytes)	U(length) followed by exactly those octets.
Seq_T(items)	U(count) followed by the items in order, each using schema <i>T</i> .
Z(z)	O(sign) U(limb_count), then that many eight-octet, little-endian limbs; canonical sign and leading-limb rules below.
Q(q)	Z(numerator) Z(denominator), with positive denominator and reduced fraction.
Name	A B byte string containing only printable ASCII octets <code>0x21..0x7e</code> ; nonempty and case sensitive. Free-form provenance is opaque B data and has no name-binding semantics.

Primitive	Encoding (continued)
ProfileId	Name(namespace) U(major_version) U(minor_version) Name(label). Complete fields form the nominal type/binding identity; they are not executable prose.

An implementation may announce finite count, width, depth and memory limits. A larger valid mathematical record is then unsupported, not truncated or partly accepted. Check lengths before allocation and retain exact count arithmetic.

A.2 Literal payloads and type tags

For integer limbs a_i in base $\beta = 2^{64}$, $z = (-1)^s \sum_{i=0}^{k-1} a_i \beta^i$, with $0 \leq a_i < \beta$. Zero is $s = 0, k = 0$; otherwise $a_{k-1} \neq 0$. A rational p/q requires $q > 0$ and $\gcd(|p|, q) = 1$; rational zero is $0/1$.

Literal type	Exact payload and admission rules
Bool	O(value), with value zero or one.
Int	Z(value), with the canonical integer form above.
Rat	Q(value), with the canonical rational form above.
Alg	Seq_Z(coefficients) U(root_index). Coefficients run from constant to highest degree. The polynomial has degree at least one, is primitive and irreducible over $\mathbb{Q}$, and has positive leading coefficient. The zero-based index selects a distinct real root in increasing order.
Bits(w)	Exactly $\lceil w/8 \rceil$ octets; bit zero is the low bit of the first octet. Unused high bits in the final octet are zero. Width is in the type, not repeated in this payload.
Dim(T,d)	Exactly the scalar payload of T ; the seven exponents already occur in its type. No extra numeric field or raw RealTerm literal is authorized by the wrapper.
Tuple, vector, matrix	Ordered Seq_Literal components with type/shape agreement; each component retains its own type.
Frame	The underlying Vec/Mat literal payload, with the complete frame annotation retained in Type; encoding and sharing cannot erase it.

An algebraic isolating interval is a witness, not an alternative value spelling. Exact factorization, minimal-polynomial selection, root isolation and selected root identification remain implementation obligations. A resultant alone does not select the intended root. Raw finite IEEE bits can be converted to their exact dyadic value; this does not recover a pre-rounding physical quantity. Signed-zero payloads remain distinct as bits but both denote rational zero. NaN and infinity have no real conversion.

Each type is U(tag) followed by the fields in this table:

Tag	Type	Remaining fields
0	Bool	None.
1	Int	None.
2	Rat	None.
3	Alg	None.

Tag	Type	Remaining fields (continued)
4	RealTerm	None.
5	Bits	$U(w)$, $w > 0$.
6	Tuple	$\text{Seq_Type}(\text{component_types})$.
7	Vec	$U(n)$ $\text{Type}(T)$, $n > 0$.
8	Mat	$U(\text{rows})$ $U(\text{cols})$ $\text{Type}(T)$; both sizes positive.
9	Fn	$\text{Seq_Type}(\text{parameter_types})$ $\text{Type}(\text{result})$.
10	Dim	$\text{Type}(\text{scalar})$, followed by exactly seven Q exponents.
11	Stream	$\text{Type}(\text{element})$.
12	Frame	$\text{Type}(\text{base})$ $\text{ProfileId}(\text{source})$, plus $\text{ProfileId}(\text{target})$ when base is Mat. Base is an unannotated Vec or Mat.

Dim accepts only an unwrapped numeric scalar; nested dimension wrappers are rejected. A function's domain obligations remain part of its interface/body. Function, stream and ungrounded real-term literals are rejected.

A framed vector's identity denotes its coordinate basis; a framed matrix's source/target denote the input/output bases of its linear map. Matching nominal identity does not prove physical alignment, origin or calibration. Axis/origin realization and uncertainty are explicit template/dependency inputs. Qualified namespaces must agree across the dependency closure.

Bare vectors/matrices are frame-free objects. Once an interface declares a frame, its annotation is mandatory. COERCE cannot add, change or remove it. Literal input may declare a frame; a computed frame change requires an explicitly typed TEMPLATE, its full body and coordinate/domain obligations. Extracting coordinates and relabelling the reassembled result is not an implicit conversion. This also governs removal of an annotation for use in geometry.

Frame-aware addition/subtraction and dot/cross require identical vector frame identities. Norm returns a scalar. A matrix $A \rightarrow B$ maps a vector in A to B through an explicit contraction template; composing $A \rightarrow B$ then $B \rightarrow C$ produces $A \rightarrow C$ in the corresponding product order. Transpose and inverse swap source/target annotations under their algebraic domains; interpreting transpose as inverse also requires orthonormality. Determinant refers to the declared coordinate matrix, without a basis-independence assertion. Projection retains its annotated source in node identity even when its result is scalar. Sharing/memoization include complete annotated types and immutable bindings.

A.3 Scopes, references and complete record order

The following fields are concatenated in the displayed order:

```

Scope = Seq_Node(nodes) Seq_Ref(exports)
Node = U(opcode) Type(result) Seq_Ref(operands) B(opcode_payload)
Literal(T) = Type(T) B(payload)

ASCII("XOPSEED1") U(1) U(0)
Seq_Dependency(dependencies)
Seq_Template(templates)
Seq_State(states)
Seq_Sample(samples)
Scope(main)
Seq_Export(exports)
Strategy(strategy)
B(provenance)

```

The magic is exactly eight ASCII octets; the two naturals specify major/minor profile (1,0). No trailing octets are allowed. Every opcode payload must decode exactly under its selected schema, with no extra fields. Empty payload means a zero-length B; the stated result type must satisfy the operator rule.

A reference is U(tag) followed by:

Tag	Reference kind	Fields and scope rule
0	Local node	U(index). A node operand refers only to an earlier node in its own scope.
1	Bound variable	U(de_bruijn_index). Zero is the nearest lexical binder.
2	Old state	U(slot); main dynamic scope only.
3	Current sample	U(slot); main dynamic scope only.
4	Dependency export	U(dependency) U(export); complete bytes and export interface must resolve.
5	Current sample raw timestamp	U(slot); main dynamic scope only. Returns the binding's exact Q(absolute_time) as Rat in the slot's source TimeConvention coordinate.
6	Current sample GPST timestamp	U(slot); main dynamic scope only. Explicitly applies the declared time mapping and returns Dim(Rat, (0, 0, 1, 0, 0, 0, 0)) in GPST SI seconds.

A scope export or state root may use any already declared node in its complete scope, but this does not authorize forward operands inside a node. Nested scopes cannot reach an outer local node through an accidental numeric index: captures enter as explicit binders. Binder order follows the payload; the last parameter has de Bruijn index zero. Substitution is capture avoiding and shifts indices beneath binders. Templates receive state/sample values as arguments.

Tags 3, 5 and 6 read the same immutable sample snapshot. Tag 5 exposes recorded source time, without asserting that a custom coordinate is SI seconds. Tag 6 preserves the registered GPST-SI rational with its time units, or invokes the complete custom mapping. Missing input and unmet/unresolved conversion domains remain explicit non-value outcomes; no offset is guessed. The SI exponent order is length, mass, time, electric current, thermodynamic temperature, amount of substance, luminous intensity.

An SI-second integrator's time arguments are declared expressions, for example

$$t = \text{SAMPLE_GPST}(j) - t_{\text{epoch}}, \quad h = \text{SAMPLE_GPST}(j) - t_{\text{previous}},$$

with matching time dimensions and GPST epoch conventions on every term. Here SAMPLE_GPST names reference tag 6, not an additional opcode. Other integration coordinates require explicit typed mappings and domain/unit hypotheses. Chronology and $h > 0$ remain recurrence obligations. Initial roots and pure template bodies exclude tags 5 and 6 as they exclude tag 3; a template receives needed timestamps as arguments. Raw ticks, phase wraps and civil-clock strings cannot silently supply these integration arguments.

```

Dependency = Name(label) O(kind) B(canonical_seed_bytes)
Template = Name(label) Seq_Type(parameters) Type(result) Scope(body)
State = Name(label) Type(T) Ref(init) Ref(next)
Sample = Name(label) Type(T) TimeConvention
         O(mode) B(binding) B(provenance)
Row = Z(logical_step) Q(absolute_time) Literal(T)
Export = Name(label) U(main_export_index) O(observation)
TimeConvention = ProfileId(identity) O(kind) B(definition)

```

Dependencies. Kind zero imports a seed value export; kind one imports a template-library seed. Content is a complete XOPSEED1 record, labels are unique, and imports are finite and acyclic. Imported values must be closed and stateless in their demanded graph. A snapshot may be closed literal/expression data; a live stream must enter through an explicit sample. Importing a seed neither executes its state machine nor silently selects a clock tick.

Templates. A body has exactly one export of its declared result type. It contains no main-state/sample references and calls only earlier local or explicitly imported templates. The encoded body fixes its meaning; a label alone does not provide an equation or authorize recursion.

State. Initial and next roots refer to the main scope. Initial roots are closed with respect to state and samples and must have proved values of the declared type. Next roots read the same old-state/input snapshot. The state table is the sole causal DELAY; all successful next values commit together. No instantaneous cycle or partial commit is defined. A required input/domain failure remains an explicit outcome.

Samples. Mode zero binding is Seq_Row(rows) with nonnegative, strictly increasing logical-step indices. Mode one binding is Name(external_port); every demanded step supplies a typed literal and absolute timestamp under the named convention. Missing rows or ports yield MISSING_INPUT. No implicit interpolation, nearest-row choice, leap-second conversion or future extrapolation is supplied. Repeated reads in one step bind the same sample. The timestamp and sample identity are retained.

Time convention. Kind zero has empty definition and exactly the registered identity ("XOP-TIME", 1, 0, "GPST-SI"). Its rational coordinate is elapsed SI seconds on continuous GPS time from 1980-01-06 00:00:00 GPST; no UTC conversion occurs. Kind one is custom, with definition U(dependency) U(template) ProfileId(target). The dependency is a complete template-library seed and the target is the registered GPST-SI identity. The selected pure template maps unambiguous rational source time to rational target time. Its mapping, interval, epoch and any leap/disambiguation tables must be in its body or complete dependencies. Ambiguous civil readings need additional binding. Repeated custom identities in the dependency closure must resolve to identical canonical definition/template bytes. Ordering and elapsed time claims require the mapping's monotonicity/domain proof, not version matching alone.

Exports. Labels are unique and observation is zero or one. Stateful equivalence preserves observation-one exports, sample identities and declared output timestamps. Even unobserved exports remain part of the ordinary value interface unless an explicit interface change is adopted.

Canonical encoding means one spelling for the same ordered scoped record, not for every mathematically equivalent expression or sharing pattern. Dependency hashes are lookup aids; complete canonical bytes establish required identity. A digest-only open manifest is not this self-contained encoding.

A.4 The one-bit plane envelope

Let C be a canonical semantic stream of positive length L . Pad it with zero octets to $512\lceil L/512 \rceil$ octets. Each block holds 64 little-endian uint64 patterns w_r . Its plane words are

$$p_c = \sum_{r=0}^{63} ((w_r \gg c) \& 1) 2^r, \quad 0 \leq c < 64.$$

```
ASCII("XOPPLAN1") U(L)
plane words: eight octets each, little endian,
              block-major then plane-major order
```

Exactly $512\lceil L/512 \rceil$ payload octets are required. Apply the same transpose to invert; reject nonzero

decoded padding, take exactly L octets, then parse XOPSEED1. Nonminimal lengths, redundant integer forms, invalid references, nonzero padding and trailing bytes have no canonical decoding. The transpose proof is in Section 1. Compression is a separate transport that must reconstruct this envelope exactly. A bit permutation does not reduce its uncompressed information content or erase high limbs.

A.5 Structural, numeric and Boolean opcodes

In the tables, S is a common underlying scalar sort reached by explicit coercions. Dimension transformations below refine the scalar signatures; matching S does not require multiplication operands to have equal units. T is a complete type. Unless stated otherwise, payload is empty. Ordinary numeric operands are strict in demanded subexpressions; IF is lazy and a lambda body is demanded only through application. A node table is a dependency order, not a command to evaluate unused branches.

ID / operator	Operands $\rightarrow$ result	Domain and payload
0 LITERAL	None $\rightarrow$ literal type	Payload is that type's canonical literal payload.
1 COERCE	Scalar $\rightarrow$ higher scalar	Only the stated injective scalar embeddings.
2 TUPLE	Ordered values $\rightarrow$ tuple	Result component types match.
3 PROJECT	Tuple/Vec/Mat $\rightarrow$ component	Payload $U(\text{index})$, in bounds. Matrix index is row-major $r \cdot \text{cols} + c$ and returns one scalar.
4 VECTOR	n scalars $\rightarrow \text{Vec}(n, S)$	Positive size; common scalar sort and dimension.
5 MATRIX	rc scalars $\rightarrow \text{Mat}(r, c, S)$	Row-major order; result fixes shape.
6 LAMBDA	Captures $\rightarrow$ Fn	Payload $\text{Seq_Type}(\text{parameters}) \text{ Scope}(\text{body})$; one body export; capture binders precede parameter binders.
7 APPLY	Function, arguments $\rightarrow$ result	Exact arity/types; substitute into the body and its domain.
8 TEMPLATE	Arguments $\rightarrow$ template result	Payload $0(\text{location}) \ U(\text{index})$. Location 0 selects local template. Location 1 makes index a dependency and adds $U(\text{imported_template_index})$.
16 ADD	$S, S \rightarrow S$	Equal dimensions.
17 SUB	$S, S \rightarrow S$	Equal dimensions.
18 NEG	$S \rightarrow S$	Numeric scalar.
19 MUL	$S, S \rightarrow S$	Add dimension vectors.
20 DIV	$S, S \rightarrow S$	Denominator nonzero; Int result forbidden, requiring explicit Rat promotion; subtract dimensions.
21 IDIV	Int, Int $\rightarrow$ Int	Nonzero divisor; Euclidean quotient.
22 MOD	Int, Int $\rightarrow$ Int	Nonzero divisor d ; $0 \leq r < d $.
23 POW_INT	$S, \text{Int} \rightarrow S$	Nonnegative exponent, or nonzero base and rational/higher result; $x^0 = 1$ by finite-product convention.
24 ABS	$S \rightarrow S$	Ordered real scalar.
25 SIGN	$S \rightarrow \text{Int}$	-1, 0, 1 according to exact sign; decision can remain unresolved.
26 MIN	$S, S \rightarrow S$	Matching dimensions; order evidence required for selection.
27 MAX	$S, S \rightarrow S$	Matching dimensions; order evidence required for selection.
28 FLOOR	Dimensionless scalar $\rightarrow$ Int	Unique n with $n \leq x < n + 1$; decision can remain unresolved.
29 CEIL	Dimensionless scalar $\rightarrow$ Int	$-\lfloor -x \rfloor$.
30 ROOT	Int n , scalar $x \rightarrow$ Alg/RealTerm	$n \geq 1$. Even n : $x \geq 0$, nonnegative root. Odd n : unique real root. Divide dimensions by n .
31 SQRT	Scalar $\rightarrow$ Alg/RealTerm	$x \geq 0$, nonnegative root; explicit scalar promotion and halved dimensions.
32 EQ	Compatible equal types $\rightarrow$ Bool	Exact value equality; no terminating general RealTerm decision is promised.

ID / operator	Operands $\rightarrow$ result	Domain and payload (continued)
33 LT	$S, S \rightarrow \text{Bool}$	Ordered scalar; matching dimensions.
34 LE	$S, S \rightarrow \text{Bool}$	Ordered scalar; matching dimensions.
35 AND_BOOL	$\text{Bool}, \text{Bool} \rightarrow \text{Bool}$	Logical conjunction, not a word operation.
36 OR_BOOL	$\text{Bool}, \text{Bool} \rightarrow \text{Bool}$	Logical disjunction.
37 NOT_BOOL	$\text{Bool} \rightarrow \text{Bool}$	Logical negation.
38 IF	$\text{Bool}, T, T \rightarrow T$	Lazy: only the selected branch must be defined. An unknown predicate retains the conditional term.
40 FULL_ADDER	$\text{Bool } a, b, c \rightarrow \text{Bool pair}$	Sum bit and carry bit, equations below.
41 FULL_SUBTRACTOR	$\text{Bool } a, b, k \rightarrow \text{Bool pair}$	Difference bit and borrow bit, equations below.
48 BIT_AND	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
49 BIT_OR	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
50 BIT_XOR	$\text{Bits}(w) \text{ pair} \rightarrow \text{Bits}(w)$	Equal explicit widths.
51 BIT_NOT	$\text{Bits}(w) \rightarrow \text{Bits}(w)$	Complement inside the declared width only.
52 BIT_SHIFT	$\text{Bits}(w), \text{Int } k \rightarrow \text{Bits}(w)$	$k \geq 0$; payload 0(direction): 0 left, 1 logical right. Shifted-out bits are deliberately discarded.
53 POPCOUNT	$\text{Bits}(w) \rightarrow \text{Int}$	Sum of the w Boolean bits.
54 BITS_TO_UINT	$\text{Bits}(w) \rightarrow \text{Int}$	$\sum_i b_i 2^i$.
55 UINT_TO_BITS	$\text{Int} \rightarrow \text{Bits}(w)$	$0 \leq x < 2^w$; no implicit modular truncation.
56 INT_SHIFT	$\text{Int}, \text{Int } k \rightarrow \text{Int}$	$k \geq 0$; payload direction 0 multiplies by 2^k , 1 takes floor division by 2^k .
57 IEEE64_TO_RAT	$\text{Bits}(64) \rightarrow \text{Rat}$	Finite IEEE binary64 pattern only; exact dyadic conversion below.
58 DECIMAL_TO_RAT	$\text{Int mantissa}, \text{Int scale} \rightarrow \text{Rat}$	Exact mantissa $\cdot 10^{-\text{scale}}$, reduced.

For IEEE conversion let s be the sign bit, e the unsigned 11-bit exponent and f the unsigned 52-bit fraction. The exact value is

$$\begin{cases} (-1)^s f 2^{-1074}, & e = 0, \\ (-1)^s (2^{52} + f) 2^{e-1075}, & 1 \leq e \leq 2046. \end{cases}$$

Exponent 2047 is outside the real conversion domain. Keep the raw operand if signed-zero or other bit provenance must remain observable. Rational operations reduce by gcd and normalize sign; algebraic operations preserve selected-root identity. The source equation-language token 1 means an all-ones word only under its separately declared language adapter.

For Boolean carry c and borrow k , the one-bit arithmetic operators are

$$\begin{aligned} s &= a \oplus b \oplus c, & c' &= (a \wedge b) \vee (c \wedge (a \oplus b)), \\ d &= a \oplus b \oplus k, & k' &= (\neg a \wedge b) \vee (k \wedge \neg(a \oplus b)), \\ a + b + c &= s + 2c', & a - b - k &= d - 2k'. \end{aligned}$$

Complements here act on one Boolean bit. Chain carry/borrow toward more significant bits, starting at zero. Multiply by placing $a_i \wedge b_j$ at position $i + j$ and adding every shifted row with retained carries. To compare padded magnitudes from high to low bits, start $(E, G) = (1, 0)$ and update simultaneously $G' = G \vee (E \wedge a \wedge \neg b)$ and $E' = E \wedge \neg(a \oplus b)$. Long division uses this exact comparison and subtraction. These are finite definitions for every finite input; no executed bit test is asserted.

A.6 Elementary, tensor, calculus and kernel opcodes

Elementary functions have the real denotations and branch conventions in Section 1. They require explicit promotion to `RealTerm`; angle results are dimensionless with declared angle convention.

ID / operator	Operands $\rightarrow$ result	Domain and payload
64 SIN	RealTerm $\rightarrow$ RealTerm	Entire sine series; dimensionless argument.
65 COS	RealTerm $\rightarrow$ RealTerm	Entire cosine series; dimensionless argument.
66 TAN	$x \rightarrow \sin x / \cos x$	$\cos x \neq 0$.
67 EXP	RealTerm $\rightarrow$ RealTerm	Entire exponential series.
68 LOG	$x \rightarrow \int_1^x du/u$	$x > 0$.
69 ATAN	$x \rightarrow \int_0^x du/(1+u^2)$	Every real x .
70 ATAN2	$y, x \rightarrow \text{angle}$	$(x, y) \neq (0, 0)$; range $(-\pi, \pi]$ with the declared quadrant cases.
71 ASIN	$x \rightarrow \text{angle}$	$-1 \leq x \leq 1$; inverse sine on $[-\pi/2, \pi/2]$.
72 ACOS	$x \rightarrow \text{angle}$	$-1 \leq x \leq 1$; inverse cosine on $[0, \pi]$.
73 PI	None $\rightarrow$ RealTerm	$4 \text{atan}(1)$.
74 POW_REAL	$x, y \rightarrow \exp(y \log x)$	$x > 0$; both arguments dimensionless. Distinct from integer powers and root branches.
80 DOT	Vec(n, S) pair $\rightarrow S$	Finite sum of pairwise products.
81 CROSS	Vec(3, S) pair $\rightarrow$ Vec(3, S)	Right-handed determinant in the declared frame.
82 NORM	Vec(n, S) $\rightarrow$ Alg/RealTerm	Nonnegative square root of dot product; retains the vector dimension.
83 TRANSPPOSE	Mat(r, c, S) $\rightarrow$ Mat(c, r, S)	Swap indices.
84 MATMUL	Mat(r, k, S), Mat(k, c, S) $\rightarrow$ Mat(r, c, S)	Finite contraction over k .
85 DET	Square matrix $\rightarrow S$	Finite signed permutation sum.
86 INVERSE	Mat(n, n, S) $\rightarrow$ Mat(n, n, S)	Nonzero determinant; scalar sort closed under division.
87 SOLVE_LINEAR	$A, b \rightarrow x, Ax = b$	Square A , nonzero determinant, compatible vector and division-capable scalar sort.
96 FIN_SUM	Int a, b , captures $\rightarrow S$	Inclusive bounds. Payload Type(S) Scope(body); index binder last. Empty interval gives additive identity.
97 FIN_PRODUCT	Int a, b , captures $\rightarrow S$	Same payload/binding; empty interval gives multiplicative identity. Terms dimensionless in this base rule.
98 DIFF	Point, captures $\rightarrow$ RealTerm	Payload Type(variable) Scope(body); variable binder last. Derivative must exist at the point.
99 INTEGRAL	a, b , captures $\rightarrow$ RealTerm	Same payload/binder; oriented finite Riemann integral. Integrability is required.
100 LIMIT	Point, captures $\rightarrow$ RealTerm	Payload Type(variable) 0(side) Scope(body). Side 0 two-sided, 1 from below, 2 from above. Unique finite limit required.
101 ODE	t_0, s_0, t , captures $\rightarrow$ state	Payload Q(interval_left) Q(interval_right) Scope(F). Closed interval includes t_0, t ; time binder then state last. Unique continuous integral-equation solution required.
102 SOLVE_ROOT	Captures $\rightarrow$ RealTerm	Payload Q(left) Q(right) Scope(f); binder x last. Exactly one real root in the open rational interval; existence and uniqueness required.
112 ASA_NA	$x, V, A, N, B \rightarrow (a, n, h, y)$	Five Bits(w) operands; result has word components a, n, y and Int h . Whole-word absorption below.
113 JK	$q, J, K, V \rightarrow \text{Bits}(w)$	Four same-width words, reading the same old q ; complement restricted to width.

Calculus binders and solution domain. Captures bind in operand order, followed by the displayed integration/differentiation variables. Each body has one export. Body and variable dimensions determine derivative/integral units. ODE supports a scalar, uniform vector, framed vector, or finite nested tuple of those dimensioned components; every scalar result leaf has sort RealTerm. Initial algebraic values are coerced componentwise while preserving all frame annotations. F has the same structure and identical nominal frame on each framed vector, with each scalar leaf dimension equal to its state dimension minus the time dimension. Rotating-frame derivatives still require the appropriate terms in F ; the type supplies none. Interval rationals use

the same declared time unit as t_0 and t .

Specifically, ODE denotes the unique continuous s on the declared closed interval I such that, componentwise,

$$s(u) = s_0 + \int_{t_0}^u F(v, s(v)) dv \quad (u \in I),$$

with a Riemann-integrable integrand along s . It is differentiable where the composite right-hand side is continuous; a derivative at a coefficient jump is not required. The derived name FLOW uses this same declaration. State-dependent discontinuities still require existence/uniqueness under the event policy. A local theorem does not certify the whole interval. General root and ODE declarations are not guaranteed solution algorithms. Unbounded sums, improper integrals, complex roots and distributional derivatives are outside these opcodes.

Dimensions. If operands have dimension vectors d_A, d_B , then MUL, DOT, CROSS and MATMUL return $d_A + d_B$; DIV returns $d_A - d_B$. NORM retains dimension. For a uniformly dimensioned $n \times n$ matrix, determinant has dimension nd_A , inverse $-d_A$, and SOLVE_LINEAR has dimension $d_b - d_A$. Addition, subtraction and order require equal dimensions. Heterogeneous Jacobians use dimensioned tuples/blocks and explicit contraction templates, not one falsely uniform matrix type. Dimensioned powers/roots must establish the annotated output dimensions from their degree at admission; a variable degree cannot contradict a fixed annotation. Dimensionless inputs avoid that degree/unit issue.

The retained kernel operators have literal equations

$$\begin{aligned} a &= x \wedge A \wedge V, & n &= a \wedge N, & h &= \text{popcount}(n \wedge B), \\ y &= \begin{cases} 0, & h > 0, \\ n, & h = 0, \end{cases} & \text{ASA_NA}(x, V, A, N, B) &= (a, n, h, y), \\ \text{JK}(q, J, K, V) &= ((J \wedge \neg_w q) \vee (\neg_w K \wedge q)) \wedge V. \end{aligned}$$

Here $\neg_w q = (2^w - 1) \oplus q$. The second mask set receives the first stage's y ; both traces remain exported when required. J, K use the same old state and final y . Absorption zeroes the whole stage output and does not silently substitute a hold rule. The state table supplies the delay.

A.7 Named templates and the proof-record interface

Named templates require explicit typed parameters and bodies. Labels alone provide no operator implementation. WRAP is $x - P[x/P]$ for $P > 0$, paired with integer winding; PHI is $(1 + \sqrt{5})/2$; FIBONACCI has initial $(0, 1)$ and delayed pair update $(a, b) \mapsto (b, a+b)$. NORMALIZE requires nonzero norm. GRADIENT/JACOBIAN assemble coordinate DIFF terms. Polynomial/Horner and Chebyshev templates preserve their finite coefficients and degrees, with $T_0 = 1, T_1 = x, T_{n+1} = 2xT_n - T_{n-1}$. Legendre, gravity, frame, third-body, pressure/shadow and station templates retain their source equations and singular/branch domains. RK4 is the finite rule in Section 1, not the continuous ODE declaration. OTAN2, both UGTS key layouts and the source geometry remain separate templates with their literal source identities. OTAN2 is not ATAN2. Modular keys do not replace absolute coordinates, time or sample bytes.

The strategy and proof records have this exact outer order:

```
Strategy = U(version) Name(proof_system) Seq_Rule(rules)
          B(priority_policy) B(cost_model) B(resource_policy)
          Seq_Proposal(history)
Rule = Name(label) B(typed_statement) B(proof_object)
Proposal = U(parent_version) U(new_version)
           B(old_canonical_graph) B(new_canonical_graph)
           B(statement) B(proof_object) B(cost_argument)
```

The named proof-system/policy profile resolves the B contents; they are not executable arbitrary source. The baseline with empty rules, policy and history admits identity-only execution strategy. XOP-R1 defines the interface, not a completed proof language or checker. Uninterpreted policy bytes confer no authority to admit a nonidentity rewrite.

For $\Gamma \vdash e \simeq e' : T$, domains under Γ and denotations on those domains must be identical. The proof interface has these rule families:

Rule family	Required premise
ASSUMPTION	The exact proposition occurs in the typed environment.
REFL, SYM, TRANS	Typed equality and compatible domain judgements.
CONGRUENCE	Equal corresponding operands, same operator/bindings and preserved definedness.
SUBSTITUTION, BETA	Capture-avoiding typed substitution; function/body domain preserved.
EXPAND	Exact catalog/template definition, including its branches and domains.
INTEGER_RING, RATIONAL_FIELD	Exact laws with every nonzero-denominator obligation retained.
ORDER, ROOT_BRANCH	Required sign, range and unique-root facts.
ANALYTIC_LEMMA	Explicit derivative/integral/limit hypotheses and proof in the accepted calculus.
STATE_SIMULATION	Initial relation, step-domain equivalence and equal declared observations for every admitted input.

A theorem identifier must resolve to an accepted theorem with discharged substitutions and premises. A named or absent checker cannot admit nonidentity proofs. Strategy versions are immutable; the previously accepted checker reviews a proposal. Replacing that checker creates a new trust boundary and cannot be certified merely by the proposal's own changed rules. Hashes identify bytes, not soundness or termination.

Memoization binds complete operator/dependency identity, arguments, sample snapshot and state step. Resource outcomes and traces also count when promised by the interface. A decreasing natural-valued measure proves termination of that rewrite phase; a finite search budget proves only bounded search. Neither establishes global optimality, speed, bounded limb growth or physical accuracy. The causal and conditional replacement proofs remain those of Section 1; no new algorithm validation is asserted here.